

A Metric Differential Privacy Mechanism for Sentence Embeddings

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Sentence embeddings represent the meaning of a given sentence using a fixed dimensional vector. Different approaches have been proposed in the Natural Language Processing (NLP) community for learning encoders that can produce accurate sentence embeddings that perform well for diverse downstream tasks requiring sentence representations. Despite prior work focusing mainly on creating accurate sentence embeddings, how to keep private the sensitive information contained in the sentences remains an unexplored research problem. In this article, we propose Covering Metric Analytic Gaussian (CMAG), a *covering* metric Differential Privacy (DP) mechanism for sentence embeddings such that minimal random noise is added to a set of sentence embeddings produced by an encoder to protect the private information expressed in those sentences. Given a sentence embedding s , CMAG considers the Mahalanobis distance between s and the other sentence embeddings s' in the local neighbourhood of s to determine the minimal amount of random noise that must be added to s to obtain provable metric DP guarantees. Experimental results show that the proposed DP mechanism protects private information better than previously proposed DP mechanisms while reporting good performance in a broad range of downstream NLP tasks.

CCS Concepts: • **Security and privacy**;

Additional Key Words and Phrases: Metric differential privacy, sentence embeddings

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1 Introduction

Billions of users worldwide interact with **Natural Language Processing (NLP)** systems such as Web search engines (e.g., Google and Bing), **Large Language Model (LLM)**-powered chat agents (e.g., ChatGPT¹ and Gemini²), and machine translation systems (e.g., DeepL³ and Google Translate⁴) on a daily basis. From the users' point of view there is no other alternative but to trust these NLP systems with their sensitive private information, such as medical conditions, financial information, and social relationships in order to obtain the information they seek. Unfortunately, even with our best efforts to prevent such scenarios, the plain text information entered to the user interfaces in the NLP systems can easily be used for purposes other than the original intentions of the users of those systems. For example, search engines determine the best advertisements to display to a user based on the user queries and to personalise the search results received by the individual users. Ad revenue is a major source of income to search engines. Indeed, it has been reported that advertisements contribute to 96% of Google's revenue.⁵ Moreover, the utterances entered into chat agents such as ChatGPT can be used to further train the underlying LLMs. Therefore, an important challenge is **how to protect the private information in texts, while retaining sufficient information for the NLP systems to provide relevant information to their users.**

To address the privacy protection challenge, we turn to **Differential Privacy (DP)** [14] mechanisms that provide theoretically guaranteed privacy protections. In DP, we add random noise sampled from a particular noise distribution to a given input such that the original information cannot be uniquely determined beyond a specified privacy level. DP mechanisms have been successfully used in various applications, such as releasing pretrained word embeddings [16] and protecting privacy when training neural networks [1]. Throughout this article, we consider a single *sentence* as the unit of text for protecting privacy. However, note that we do not impose any length constraints on the sentences considered. From the viewpoint of practical applicability, we identify the following two desirable properties that must be satisfied by a DP mechanism when protecting user privacy in sentences.

Property 1: Any machine learning model that is already trained on sentence embeddings without using any random noise does not have to be retrained to work with noise-added sentence embeddings.

In this article, we do not require the machine learning models, such as text classifiers that are already trained on embedded texts, to be retrained when they are provided with privatised sentence embeddings that have random noise being added to the embeddings. Training machine learning models such as LLMs is costly, and having to retrain such models each time when we have added some random noise to the input embeddings is not feasible in practice.

Property 2: Only the minimal amount of random noise that provides user-specified privacy guarantees must be added to the sentence embeddings.

Adding random noise to sentence embeddings is a process that requires balancing a subtle trade-off between protecting privacy and preserving as much semantic information in the embeddings as possible to obtain satisfactory performance in downstream NLP tasks that use those sentence embeddings. Adding too much noise to sentence embeddings would result in poor performance in downstream NLP tasks.

¹<https://chat.openai.com/>

²<https://gemini.google.com/app>

³<https://www.deepl.com/>

⁴<https://translate.google.com/>

⁵<https://www.wordstream.com/articles/google-earnings>

In this article, we propose CMAG, a metric DP [21] mechanism for privatising sentence embeddings by adding random noise sampled from a Gaussian distribution that satisfies the above-mentioned properties. Given a set of sentence embeddings, CMAG determines the minimum amount of random noise that must be added to each sentence embedding such that a given sentence is indistinguishable from its nearest neighbour sentences in the embedding space. The noise-added sentence embeddings can then be used in machine learning models trained for downstream NLP tasks *without* any retraining of those models, thus satisfying the first property as stated above. Instead of determining a single global noise level for all sentences, CMAG calculates the noise parameters per each sentence such that a user-specified privacy level ϵ is guaranteed. By doing so, CMAG aims to satisfy Property 2.

Unlike most prior proposals for privatising text representations, CMAG has theoretically proven privacy guarantees. Moreover, it does not require modifying or retraining the sentence encoder that is being used to produce the sentence embeddings, and applies randomly sampled noise *after* the sentence embeddings in a post-processing manner. Therefore, in theory, CMAG can be used with any existing sentence encoder, which enables us to use **state-of-the-art (SoTA)** sentence encoders that are based on contrastive learning methods [42]. Learning sentence embeddings is an active research field with more accurate and efficient sentence encoders being developed in a continuous manner [25, 35, 41]. The ability to privatise sentence embeddings that are produced using any sentence encoder without having to retrain the encoder or the downstream NLP model that uses those sentence embeddings can be considered as an attractive property of CMAG.

We compare CMAG against previously proposed DP mechanisms in both *privacy* (Section 4.3) and *utility* (Section 4.4) experiments. First, in the privacy experiments, we evaluate the level of privacy protected by CMAG against previously proposed DP mechanisms. A sentence encoder can be seen as a mapping between sentences (which can be seen as a discrete set of words) and points in a high-dimensional continuous vector space. Therefore, even if we add random noise to a sentence embedding, a nearest neighbour attack could easily return the original embedding for the corresponding sentence because there will be no other actual sentence in the embedding space near the noise-added target sentence embedding. Therefore, it is important to verify whether one can guess the information in a sentence using its noise-added sentence embedding according to a particular privacy mechanism. Experimental results from our privacy evaluations show that CMAG guarantees similar (or better) privacy levels with the same level of random noise compared with the previously proposed DP mechanisms for text representations.

Second, in the utility experiments, we evaluate any degradation in downstream task performance when the noise-added sentence embeddings produced by CMAG are used as the input. For this purpose, we use a diverse set of NLP tasks that have been frequently used in prior work to evaluate the utility of the privatised sentence embeddings [8]. Specifically, we use the following tasks: subjectivity status detection [32], sentiment classification [36], question type classification [27], natural language inference [30], paraphrase detection [13], semantic textual similarity [10], product review classification [39], and movie review classification [39]. In most of those tasks, CMAG outperforms previously proposed privacy mechanisms when applied to sentence embeddings, establishing a novel SoTA. Source code implementation to reproduce the findings reported in this article will be publicly released upon acceptance of the article.

2 Related Work

Learning models from data with DP guarantees has been studied under *private learning* [24]. Abadi et al. [1] proposed a DP stochastic gradient descent by adding Gaussian noise to the gradient of the loss function. Rogers et al. [34] combined multiple DP algorithms using adaptive parameters.

However, compared with continuous input spaces, such as in computer vision [44], DP mechanisms for discrete inputs such as text remain understudied.

Wang et al. [40] proposed *WordDP* to achieve certified robustness against word substitution attacks in text classification. However, WordDP does *not* seek DP protection for the training data as we consider here, and uses DP randomness for certified robustness during inference time with respect to a test input. Krishna et al. [26] proposed AdePT, an autoencoder-based approach to generate differentially private text transformations. However, Habernal [19] showed that AdePT is *not* differentially private, as claimed by Krishna et al. [26].

DPText [2, 6] was proposed as a DP-based noise addition mechanism that consists of an autoencoder for text representations and private attribute discriminators. However, Habernal [20] later showed that the text embeddings produced by DPText are not differentially private and emphasised the importance of designing theoretically sound DP mechanisms. These prior repeated false claims of DP of privatisation mechanisms show the difficulty of developing theoretically provable and empirically performant DP mechanisms. On the other hand, as we show later in this article, CMAG has theoretically provable DP guarantees.

A popular technique to protect textual information of a user from getting exposed to a malicious third party is to use encryption. Only the intended party can decrypt the encrypted text using a secret key that is being shared. However, encryption methods distort the information present in a sentence such that it cannot be predicted without first applying the corresponding decryption. Therefore, encrypted texts cannot be directly handled by machine learning models that are trained on sentence embeddings. The majority of the NLP systems that we described in Section 1 require sentence embeddings computed from non-encrypted plain text inputs. Consequently, we do not consider encryption-based methods in this article.

Lyu et al. [29] proposed **DP Neural Representations (DPNRs)** to preserve the privacy of text representations by first randomly masking words from the input texts and then adding Laplacian noise. DPNR uses a global, neighbourhood-insensitive fixed Laplacian noise distribution. On the other hand, CMAG varies the noise it applies to each sentence embedding, which enables it to utilise a fixed noise budget more effectively. Feyisetan et al. [15] proposed a DP mechanism in which they first add Laplacian noise to word embeddings and then return the nearest neighbour to the noise-added embedding as the output. However, the ℓ_2 norm of the noise vector scales almost linearly with the dimensionality of the embedding space. To address this issue, Feyisetan and Kasiviswanathan [16] projected the word embeddings to a lower-dimensional space before adding Laplacian noise.

Xu et al. [43] proposed the Mahalanobis DP mechanism, which adds elliptical noise considering the covariance matrix computed from the word embeddings. Unlike the Gaussian or Laplacian mechanisms, the Mahalanobis mechanism adds heterogeneous noise along different directions such that the words in sparse regions in the embedding space have a sufficient likelihood of replacement without sacrificing the overall utility. They show the Mahalanobis mechanism to be superior to the Laplacian mechanism for protecting privacy in word embeddings. The Mahalanobis mechanism is a special instance of metric (Lipschitz) DP originated in privacy-preserving geolocation studies [3], in which Euclidean distance was used as the distance metric. However, Xu et al. [43] evaluate their method only for static word embeddings and did not consider sentence embeddings. Li et al. [28] used adversarial learning to build NLP models such as **part-of-speech (PoS)** taggers that cannot predict the writer's age or sex while accurately predicting the PoS tags. Despite their empirical success, this approach does not have any formal DP guarantees. Moreover, their method is limited to static word embeddings.

Bollegala et al. [8] proposed **Neighbourhood Aware Differential Privacy (NADP)** for static word embeddings, in which they first create a nearest neighbour graph for a set of word

embeddings. Words are represented by the vertices of this graph, whereas two words that are mutual nearest neighbours (determined according to the Jaccard similarity computed over their neighbourhood sets) are connected by the edges. They add random Gaussian noise to word embeddings such that a word can be *hidden* among its neighbouring words in the embedding space. However, compared with the set of static word embeddings covering a predefined vocabulary of words, the set of sentences that can be generated from the same vocabulary is infinitely large because words can be sampled with repetition to generate sentences. Therefore, NADP cannot be directly extended to sentence embeddings produced by **masked language models (MLMs)**.

Most prior applications of DP to text representations focus on word embeddings [8, 15, 28, 43] or do not provide provable DP guarantees when applied to text embeddings [2, 6]. In contrast, we focus on the problem of developing DP mechanisms for sentence embeddings produced by pretrained sentence encoders. Gao et al. [17] proposed SimCSE, a contrastive learning-based method for learning accurate sentence embeddings. SimCSE adds dropout noise to synthetically generate similar sentences (positive training instances) to design an unsupervised sentence-embedding learning approach. Next, they further improve the sentence embeddings via a supervised approach in which they use annotated sentence pairs from natural language inference datasets. This approach has been further extended by follow up work, advancing the SoTA for sentence embeddings [25, 35, 41]. We use the original SimCSE model by Gao et al. [17] later in our experiments to obtain sentence embeddings.

TopicDP, proposed by the authors of [38], is a method specifically designed to protect privacy in topic mining and is not a DP mechanism for sentence embeddings, which is the focus of our article. TopicDP requires a topic modelling method such as **Latent Dirichlet Allocation (LDA)** [7] to compute the topic versus document matrix, in which elements indicate the probabilities, $p(t_i|d_j)$, that the document d_j belongs to topic t_i . They then add random Gaussian noise to these probabilities to make the topic associations private. On the other hand, in this article, we consider the problem of protecting the sensitive information encoded in sentence embeddings. More importantly, TopicDP does not use sentence embeddings, which makes a direct comparison between TopicDP and our proposed CMAG impossible. Note that one can apply, for example, a clustering method such as k -means clustering to derive topics/clusters from a set of noise-added sentence embeddings. Therefore, our proposed DP mechanism, CMAG, can be used in principle to ensure privacy in topic modelling as well. However, this is an extended application of CMAG, which we do not explore in this article.

3 Private Sentence Embeddings

In this section, we will develop a provable DP mechanism for sentence embeddings. For this purpose, in Section 3.2, we will first provide formal definitions for classical and metric DP to be used in the subsequent sections. Next, in Section 3.3, we will extend the notion of DP for the *covering metric* case, which has special relevance to sentence embeddings. We will list the theoretical findings in the form of Theorems and provide their proofs in the Appendix.

3.1 Threat Model

Before we discuss the details of the proposed DP mechanisms for sentence embeddings, let us describe the threat model considered in this article, which is illustrated in Figure 1. Our goal in this article is to protect the privacy of the users accessing an NLP system via textual inputs. The textual inputs sent to the external NLP applications such as chatbots or search engines can contain sensitive information about the users. Although the users might trust the external NLP application providers to use their sensitive information for the purpose of providing the contracted service to them, it is not only these external service providers that have access to their sensitive information

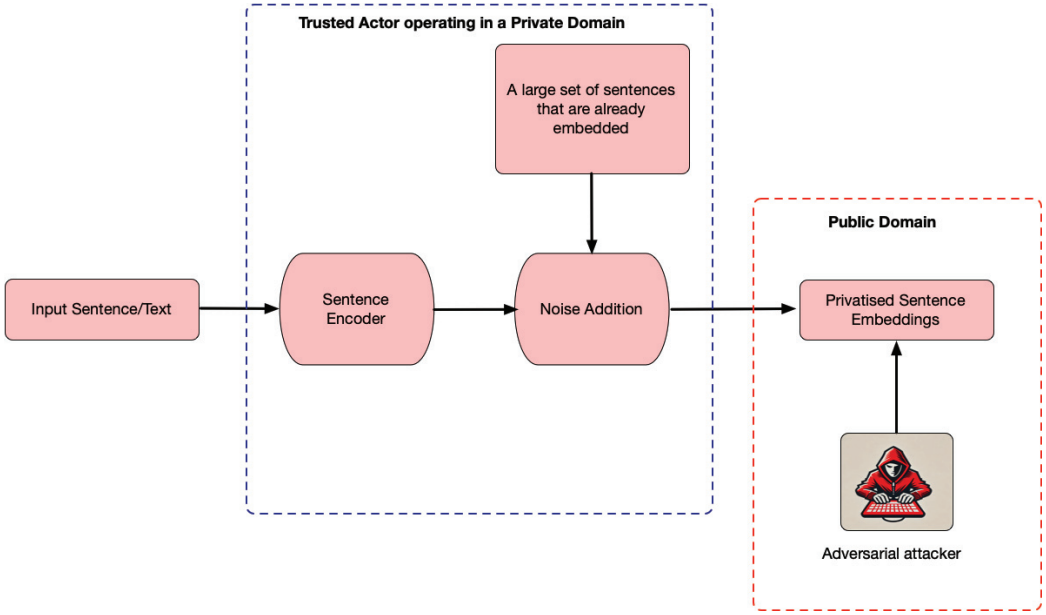


Fig. 1. The illustration shows the threat model considered in this article. A sentence containing sensitive information that should not be exposed to external parties is given as the input to a trusted agent. The trusted agent then embeds the given input sentence using a sentence encoder and finds similar sentences that are already embedded to determine the level of random noise that needs to be added to the embedded sentence. The sentence encoder and the corpus from which the nearest neighbours are selected are in a secure private domain, not externally accessible. The noise-added sentence embeddings are then provided to external parties who would like to use those sentence embeddings to train NLP models or perform inferences using pretrained NLP models. Adversarial attackers are also operating on this public domain and can get access to the noise-added sentence embeddings. They might carry out various attacks, such as denoising operations, hoping to discover the sensitive information in the original input sentences.

but also adversaries with malicious intents. For example, an adversary might want to train an LLM using the sensitive information about the users to develop an ad recommendation system without obtaining prior consent from the users. We would like to prevent such adversaries from discovering the sensitive information about the users even in the case in which they did somehow manage to get access to the sentence embeddings.

We assume that the users are willing to trust an agent that would act on their behalf to privatise the sentence embeddings before sending those sentences to external NLP systems. This trusted agent could be, for example, a software tool that resides in their own computer systems within a private domain, inaccessible from the outside. A sentence containing sensitive information that should not be exposed to external parties is given as the input to the trusted agent. The trusted agent then embeds the given input sentence using a sentence encoder and finds similar sentences that are already embedded to determine the level of random noise to be added to the embedded sentence. The sentence encoder and the corpus from which the nearest neighbours are selected are in a secure private domain, not externally accessible. This sentence corpus could also be specific to each user, such as a set of selected sentences from their past emails. The noise-added sentence embeddings are then provided to external parties who would like to use those sentence embeddings to train NLP models or perform inferences using pretrained NLP models. Adversarial attackers are also operating in this public domain and can get access to the noise-added sentence

embeddings. They might carry out various attacks, such as denoising operations, hoping to discover the sensitive information in the original input sentences.

3.2 Metric Differential Privacy

Let us assume that we are interested in privatising a given set $\mathcal{X}$ of sentences. Each sentence s in $\mathcal{X}$ consists of a sequence of words. We can represent s by an n -dimensional vector (aka embedding) s in a real-valued embedding space $\mathbb{R}^n$ (i.e., $s \in \mathbb{R}^n$), using some pre-trained sentence encoder. A random mechanism M that adds noise to the sentence embeddings is said to be differentially private if it satisfies Definition 1.

Definition 1 ((ϵ, δ)-Differential Privacy). Let ϵ, δ be real numbers such that $\epsilon \geq 0, \delta \in [0, 1]$. A randomised algorithm $M : \mathcal{X} \rightarrow \mathbb{R}^n$ satisfies (ϵ, δ)-DP with $\epsilon \geq 0$ and $\delta \in [0, 1]$ if for any inputs $s, s' \in \mathcal{X}$ and for any measurable set $E \subset \mathbb{R}^n$ the following holds:

$$\Pr\{M(s) \in E\} \leq e^\epsilon \Pr\{M(s') \in E\} + \delta. \quad (3.1)$$

Here, ϵ represents the level of privacy ensured by M , where smaller ϵ values result in stronger privacy guarantees. Although numerous types of (ϵ, δ)-DP mechanisms have been suggested in the literature, the classical Gaussian mechanism is arguably the most widely used mechanism in applications. However, in its original setting, classical Gaussian mechanism can only work in the case $\epsilon \in (0, 1)$. To overcome this restriction, Balle and Wang suggested the **Analytic Gaussian (AG) Mechanism** [5, Theorem 8, Theorem 9], whose modified version will be briefly reviewed here. For details of this modified version, see [8, Appendix C.1]. First, let ϵ and δ be real numbers such that $\epsilon \geq 0$ and $\delta \in (0, 1)$. Let $\simeq$ be a symmetric neighbouring relation on $\mathcal{X}$ and we denote by $\|s\|$ the Euclidean norm of $s \in \mathcal{X}$. We also denote by $\Phi(t)$ the **Cumulative Density Function (CDF)** of the standard univariate Gaussian distribution:

$$\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-y^2/2} dy$$

and put

$$g(u) = \Phi\left(\frac{1}{2u} - \epsilon u\right) - e^\epsilon \Phi\left(-\frac{1}{2u} - \epsilon u\right).$$

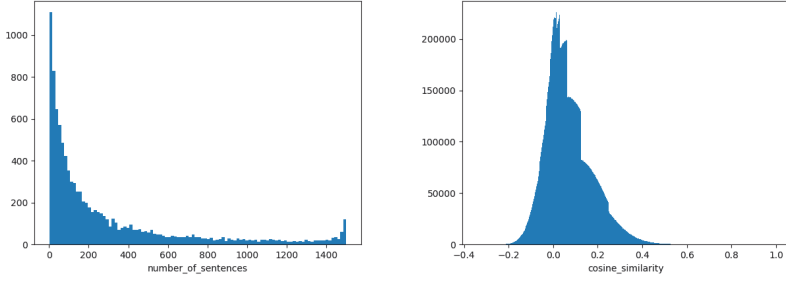
It is proven in [8, Appendix C.1] that $g(u) = \delta$ has a unique real solution u^* .

Definition 2 (Analytic Gaussian Mechanism). Put $\Delta = \sup_{s \simeq s'} \|s - s'\|$ and $\sigma = u^* \Delta$. The Analytic Gaussian (AG) mechanism M produces a noise-added embedding $M(s)$ for a sentence embedding s , where $M(s) = s + z$ with $z \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$.

Theorem 1 by Balle and Wang [5] states that the AG mechanism is a DP mechanism.

THEOREM 1. *The Analytic Gaussian (AG) mechanism given by $M(s) = s + z$ with $z \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$ satisfies (ϵ, δ)-DP.*

As seen in AG, one limitation of the classical DP (given by Definition 1) is that the level of privacy is determined purely by the privacy parameter ϵ ($+\delta$) and does not take into consideration the *distance* between the items in the metric space. This is particularly problematic in application domains such as text embeddings because even if two sentences contain different sets of words, they can still express the same meaning, as in the case of paraphrases. For example, the two sentences “The cat sat on the rug” and “The feline rested on the mat” use different sets of words but express almost the same meaning. Consequently, an accurate sentence encoder would represent those two sentences with two vectors that are close to each other in the embedding space. Therefore, it is



(a) Histogram for the number of nearest neighbours for a sentence. (b) Histogram for the cosine similarity between a pair of sentences.

Fig. 2. The two figures show the locality and sparsity of sentence embedding neighbourhoods. Figure 2(a) shows the histogram for the number of sentences that appear within a radius of 14 (i.e., neighbourhood) measured using the Euclidean distance to each target sentence. We see that most sentences have smaller neighbourhoods. Figure 2(b) shows the histogram of cosine similarities between pairs of sentences, computed using their sentence embeddings. We see a clear peak around 0, indicating that most sentence pairs are dissimilar. Note that the locality and sparsity properties described here are not required for the mathematical proofs presented later in the article.

important to consider the distance $d(s, s')$ (measured using some distance metric d) between s and s' in the embedding space when measuring the level of privacy guaranteed by a DP mechanism. Metric Differential Privacy defined by Definition 3 provides such a variant of DP that incorporates the distance between instances.

Definition 3 ((ϵ, δ)-Metric Differential Privacy). Let $(\mathcal{X}, d)$ be a metric space and ϵ, δ be real numbers such that $\epsilon \geq 0, \delta \in [0, 1]$. A randomised algorithm $M : \mathcal{X} \rightarrow \mathbb{R}^n$ satisfies (ϵ, δ) - $d_{\mathcal{X}}$ -privacy if for any $s, s' \in \mathcal{X}$ and for any measurable set $E \subset \mathbb{R}^n$, the following holds:

$$\Pr\{M(s) \in E\} \leq e^{\epsilon d(s, s')} \Pr\{M(s') \in E\} + \delta. \quad (3.2)$$

Metric DP ($d_{\mathcal{X}}$ -privacy) originated in privacy-preserving geolocation studies [3, 11], in which Euclidean distance has been frequently used as the preferred distance metric. Xu et al. [43] applied metric-DP for static word embeddings using Mahalanobis distance.

3.3 Covering Metric Differential Privacy

Recall that a sentence encoder h can be seen as a function that maps a discrete object (i.e., a sentence that contains a sequence of words, sampled from a vocabulary) to a continuous embedding. However, not all sequences of words form valid sentences. Therefore, there might not necessarily be a corresponding sentence for each point in the embedding space. When we measure the distance between sentences using some distance metric (e.g., Euclidean distance) in the embedding space and compute the nearest neighbours for each sentence, we obtain highly sparse neighbourhoods, as shown in Figure 2. Moreover, the number of neighbours that are within a fixed distance from a sentence (which we refer to as the *density* of the neighbourhood of a sentence) differs from one sentence to another. This is because some sentences can be paraphrased in various different ways, resulting in many similar sentences, whereas other sentences might not have semantically similar variants even in a large corpus. As we later see in empirical experiments, these properties of sentence embedding spaces determine the performance of DP mechanisms.

Therefore, in order to successfully apply the concept of metric-DP for sentence embeddings, we must consider the density of the neighbourhood of a sentence. To understand this further, let us consider the scenario in which we would like to add random noise to a set of sentence embeddings to protect the information encoded in those sentence embeddings. First, let $N(s_i)$ be the number of neighbours of a sentence $s_i \in \mathcal{X}$ that lies within a ball of radius $r(\geq 0)$ (i.e., $N(s_i) = |\{s \in \mathcal{X} | d(s_i, s) \leq r\}|$). By adding some random noise to the sentence embedding s_i , we can make it appear to be one of its neighbours. If there are many such neighbours (i.e., $N(s_i)$ is large), we would need to add only a small amount of random noise to s_i to achieve this objective. On the other hand, in a sparse neighbourhood (i.e., when $N(s_i)$ is small), we would need to add sufficiently large noise to s_i because, otherwise, the closest neighbour to the noise-added version of s_i will still be itself. Although adding a high level of noise to all sentences would improve the privacy of those sentences, it would likely degrade the performance in downstream applications that use the noise-added sentence embeddings. Therefore, we must consider the neighbourhood of each sentence in the set of sentences given to privatise when deciding the amount of random noise to be added to each sentence embedding. We extend the concept of metric-DP to sets of objects, covering different targets as given by Definition 4, which we refer to as the Covering Metric DP.

Definition 4 (Covering Metric Differential Privacy). Let $\{(\mathcal{X}_\lambda, d_\lambda)\}_{\lambda \in \Lambda}$ be a collection of metric spaces such that for any pair $\mathcal{X}_\lambda, \mathcal{X}_\mu$ and any element $s, s' \in \mathcal{X}_\lambda \cap \mathcal{X}_\mu$, we have that $d_\lambda(s, s') = d_\mu(s, s')$. For any $\varepsilon \geq 0$ and $\delta_\lambda \in [0, 1]$ ($\lambda \in \Lambda$), a randomised algorithm $M : \mathcal{X} = \bigcup_{\lambda \in \Lambda} \mathcal{X}_\lambda \rightarrow \mathbb{R}^n$ satisfies $(\varepsilon, \{\delta_\lambda\}_{\lambda \in \Lambda})$ - $d_\mathcal{X}$ -privacy with respect to $\{(\mathcal{X}_\lambda, d_\lambda)\}_{\lambda \in \Lambda}$ if for any $\lambda \in \Lambda$ and any $s, s' \in \mathcal{X}_\lambda$ and any measurable set $E \subset \mathbb{R}^n$, the following holds:

$$\Pr\{M(s) \in E\} \leq e^{\varepsilon d_\lambda(s, s')} \Pr\{M(s') \in E\} + \delta_\lambda.$$

As stated in Definition 4, covering metric DP is defined over a collection of metric spaces, each consisting of its own distance metric, d_λ . This is in contrast to the definition of metric DP (given in Definition 1), which has a single distance metric measured over all inputs. In other words, covering metric DP requires that the mechanism protects the privacy of each sentence not only from the global point of view but also taking into account its local (i.e., neighbourhood's) information. Moreover, note that in covering metric DP we have different δ_λ associated with each metric space such that the same ε level of privacy is guaranteed for all of the inputs. This modification enables us to incorporate the local information of a sentence into the mechanism used to perturb its sentence embedding.

As we discussed previously, the densities of the neighbourhoods of sentence embeddings tend to be both diverse and sparse compared with that of word embeddings. Therefore, in covering metric DP, we compute the distance metrics as the Mahalanobis distance determined by the inverse of the scaled sample covariance matrix corresponding to the sentence embeddings in the neighbourhoods of each sentence to be privatised. In general, let $(\mathcal{X}, d_\Sigma)$ be a metric space equipped with the Mahalanobis distance (given by Equation (3.3)) d_Σ , where Σ is a positive definite matrix and for any $s_1, s_2 \in \mathcal{X}$ with embeddings $\mathbf{s}_1 = (s_{11}, s_{21}, \dots, s_{n1})^\top$, $\mathbf{s}_2 = (s_{12}, s_{22}, \dots, s_{n2})^\top \in \mathbb{R}^n$,

$$d_\Sigma(s_1, s_2) = \sqrt{(\mathbf{s}_1 - \mathbf{s}_2)^\top \Sigma^{-1} (\mathbf{s}_1 - \mathbf{s}_2)}. \quad (3.3)$$

We put $\mathbf{U} = \Sigma^{1/2}$, that is, $\mathbf{U}$ is the unique positive definite matrix satisfying $\mathbf{U}^2 = \mathbf{U}\mathbf{U}^\top = \Sigma$. Here, suppose that $\mathcal{X}$ consists of m sentences $\mathcal{X} = \{s_1, \dots, s_m\}$ with embeddings $\mathbf{s}_j = (s_{1j}, s_{2j}, \dots, s_{nj})^\top \in \mathbb{R}^n$ ($j = 1, \dots, m$) and put $\bar{\mathbf{s}} = \frac{1}{m} \sum_{j=1}^m \mathbf{s}_j$. Then, we denote by $V_\mathcal{X} = (v_{ij})_{1 \leq i, j \leq n}$ the sample covariance matrix corresponding to these sentence embeddings:

ALGORITHM 1: Construction of the Positive Definite Sample Covariance Matrix.

-
- 1: **Inputs** : Sentences $\mathcal{X} = \{s_1, \dots, s_m\}$ with embeddings $\mathbf{X} = \{s_1, \dots, s_m\}$
 - 2: **Outputs** : Positive definite sample covariance matrix Σ
 - 3: Define $V = \frac{1}{m-1} \sum_{j=1}^m (s_j - \bar{s})(s_j - \bar{s})^\top$
 - 4: Define $r = \text{rank}(V)$
 - 5: If $r < n$, then
 - 6: Diagonalise V such that $D = (d_{ij})_{1 \leq i, j \leq n} = P^\top V P$ satisfies $d_{ii} = 0$ ($r+1 \leq i \leq n$)
 - 7: Define $V = P(D + D_{r, 10^{-7}})P^\top$
 - 8: Compute $\Sigma = \frac{n}{\text{tr}(V)} V$
 - 9: Return Σ
-

$$V_{\mathcal{X}} = \frac{1}{m-1} \sum_{j=1}^m (s_j - \bar{s})(s_j - \bar{s})^\top.$$

As stated above, the matrix Σ is taken to be the sample covariance matrix scaled by the mean of sample variances $\bar{v}$:

$$\Sigma = \frac{1}{\bar{v}} V_{\mathcal{X}} \quad \left(\bar{v} = \frac{\text{tr}(V_{\mathcal{X}})}{n} = \frac{v_{11} + \dots + v_{nn}}{n} \right). \quad (3.4)$$

By definition, Σ is an $n \times n$ positive semi-definite symmetric matrix and $\text{tr}(\Sigma) = n = \text{tr}(\mathbf{I}_n)$. It is known that the matrix rank of Σ is at most $m-1$. Therefore, when the sample size m is too small, Σ can be singular, in which case we need a minor modification of Σ to be non-singular. Note that when $r = \text{rank}(\Sigma) < n$, there exists an orthogonal matrix P and the diagonal matrix $D = (d_{ij})_{1 \leq i, j \leq n}$ such that $D = P^\top \Sigma P$ and $d_{ii} = 0$ ($r+1 \leq i \leq n$). Thus, by putting $D_{r, \gamma} = (d'_{ij})_{1 \leq i, j \leq n}$, where

$$d'_{ij} = \begin{cases} \gamma & (i = j, r+1 \leq i \leq n) \\ 0 & (\text{otherwise}) \end{cases}$$

for sufficiently small γ , the symmetric matrix $\Sigma' = P(D + D_{r, \gamma})P^\top$ becomes positive definite and the two matrices Σ and Σ' are almost the same. In this article, we set $\gamma = 10^{-7}$ for numerical computational reasons. Algorithm 1 presents the steps to construct the positive definite sample covariance matrix for a given set of sentence embeddings.

Different mechanisms have been proposed in prior work on DP to add random noise to an object to guarantee its privacy, such as the Laplace mechanism (which samples noise from a Laplacian distribution) and Gaussian mechanism (which samples noise from a Gaussian distribution). In our case, we are interested in adding random noise to embedding vectors. Prior work on text embedding learning [4, 31, 33] has shown that, for example, static word embeddings are learned in the ℓ_2 space, and their squared ℓ_2 norm is linearly related to the log-frequency of the word in the training corpus [4]. Bollegala et al. [9] extended this result to n -grams of words and proved that this property holds for phrases as well. Therefore, it is natural to consider Gaussian noise, which corresponds to the ℓ_2 embedding space used by many static word-embedding learning methods rather than Laplacian noise, which relates to the ℓ_1 norm. For this purpose, we use the analytical Gaussian mechanism proposed by Balle and Wang [5] in this article.

Let $\mathcal{X}^{od}$ be the set of all off-diagonal elements of the Cartesian product $\mathcal{X} \times \mathcal{X}$, that is, $\mathcal{X}^{od} = \{(s, s') \in \mathcal{X} \times \mathcal{X} \mid s \neq s'\}$. Then, we define

$$f(d, \sigma) := \Phi\left(\frac{d}{2\sigma} - \varepsilon\sigma\right) - e^{\varepsilon d} \Phi\left(-\frac{d}{2\sigma} - \varepsilon\sigma\right) \quad (d > 0, \sigma > 0)$$

and for any $(s, s') \in \mathcal{X}^{od}$, we put $f_{s, s'}(\sigma) = f(d_\Sigma(s, s'), \sigma)$.

ALGORITHM 2: MAG.

```

1: Inputs : Sentences  $\mathcal{X} = \{s_1, \dots, s_m\}$  with embeddings  $X = \{s_1, \dots, s_m\}$ ,  $\varepsilon \geq 0$ ,  $\delta \in (0, 1)$ 
2: Outputs : Perturbed sentence embeddings  $\hat{X} = \{\hat{s}_1, \dots, \hat{s}_m\}$ 
3: Define  $\hat{X} = \{\}$ 
4: Compute the positive definite sample covariance matrix  $\Sigma$  from  $X$  by using Algorithm 1
5: Define  $\mathbf{U} = \Sigma^{1/2}$ 
6: Compute  $d_0 = \max\{d_\Sigma(s, s') \mid (s, s') \in \mathcal{X}^{od}\}$ 
7: Compute the real solution  $\sigma_0$  of the equation  $f(d_0, \sigma) = \delta$ 
8: For  $i = 1, \dots, m$  do
9:   sample  $\mathbf{z} \sim N(0, \sigma_0^2 \mathbf{I}_n)$ 
10:   $\hat{s}_i = s_i + \mathbf{U}\mathbf{z}$ 
11:   $\hat{X} = \hat{X} + \{\hat{s}_i\}$ 
12: Return  $\hat{X}$ 

```

Theorem 2 states a condition under which the Gaussian mechanism satisfies (ε, δ) - d_X -privacy. Our next aim is to give an efficient algorithm to find the minimum σ satisfying this condition.

THEOREM 2. *The Gaussian output perturbation mechanism $M(s) = s + \mathbf{U}\mathbf{z}$ ($s \in \mathcal{X}$) with $\mathbf{z} \sim N(0, \sigma^2 \mathbf{I}_n)$ satisfies (ε, δ) - d_X -privacy if and only if $f_{s,s'}(\sigma) \leq \delta$ for any pair $(s, s') \in \mathcal{X}^{od}$.*

PROOF. The proof of Theorem 2 is given in Appendix A.1. □

Theorem 3 proves several properties of the function f that will become important when determining the noise level that would guarantee a user-specified privacy parameter ε .

THEOREM 3.

- (1) For any pair $(s, s') \in \mathcal{X}^{od}$, $f_{s,s'}(\sigma)$ is strictly decreasing on $(0, \infty)$, with $\lim_{\sigma \rightarrow +0} f_{s,s'}(\sigma) = 1$, and $\lim_{\sigma \rightarrow \infty} f_{s,s'}(\sigma) = 0$.
- (2) For any pair $(s, s') \in \mathcal{X}^{od}$ and $\delta \in (0, 1)$, the equation $f_{s,s'}(\sigma) = \delta$ has the unique real solution $\sigma(s, s', \delta)$, which satisfies $\sigma(s, s', \delta) = \inf\{\sigma \mid f_{s,s'}(\sigma) \leq \delta\}$.
- (3) Put $\sigma_0 = \max\{\sigma(s, s', \delta) \mid (s, s') \in \mathcal{X}^{od}\}$. The Gaussian output perturbation mechanism $M(s) = s + \mathbf{U}\mathbf{z}$ ($s \in \mathcal{X}$) with $\mathbf{z} \sim N(0, \sigma_0^2 \mathbf{I}_n)$ satisfies (ε, δ) - d_X -privacy.
- (4) Put $(s_0, s_0') = \operatorname{argmax}_{(s,s') \in \mathcal{X}^{od}} d_\Sigma(s, s')$. Then, we have that $\sigma_0 = \max\{\sigma(s, s', \delta) \mid (s, s') \in \mathcal{X}^{od}\} = \sigma(s_0, s_0', \delta)$.

PROOF. The proof of Theorem 3 is given in Appendix A.2. □

Remark 1. Hereafter, we denote by $\sigma(\mathcal{X}, \delta)$ the number σ_0 of Theorem 3 (4).

Theorems 2 and 3 collectively give us a noise addition method, which we call the **Metric Analytic Gaussian (MAG)** mechanism.

Definition 5 (Metric Analytical Gaussian Mechanism). Let $\sigma_0 = \sigma(\mathcal{X}, \delta)$ be the unique real solution of the equation $f_{s_0, s_0'}(\sigma) = \delta$, where $(s_0, s_0') = \operatorname{argmax}_{(s,s') \in \mathcal{X}^{od}} d_\Sigma(s, s')$. The MAG mechanism M produces a noise-added embedding $M(s)$ for a sentence embedding s , where $M(s) = s + \mathbf{U}\mathbf{z}$ with $\mathbf{z} \sim N(0, \sigma_0^2 \mathbf{I}_n)$.

Theorem 3 guarantees that MAG satisfies (ε, δ) - d_X -privacy. Algorithm 2 gives us one way to implement MAG.

Let $\{(\mathcal{X}_j, d_{\Sigma_j})\}_{j \in J}$ be a collection of metric spaces with Mahalanobis distances d_{Σ_j} such that $\Sigma_i = \Sigma_j$ whenever $\mathcal{X}_i \cap \mathcal{X}_j \neq \emptyset$ and $\delta_j \in [0, 1]$ ($j \in J$). Put $\mathcal{X} = \bigcup_{j \in J} \mathcal{X}_j$ and $\mathbf{U}_j = \Sigma_j^{1/2}$ ($j \in J$). Let $G = G(\mathcal{X})$ be a graph with vertices $\mathcal{V}(G) = \{\mathcal{X}_j\}_{j \in J}$ and edges $\mathcal{E}(G) = \{(\mathcal{X}_i, \mathcal{X}_j) \mid \mathcal{X}_i \cap \mathcal{X}_j \neq \emptyset\}$.

ALGORITHM 3: Construction of the Graph $G(\mathcal{X})$ and the Connected Components $C(\mathcal{X})$.

```

1: Inputs : A covering of sentences  $\mathcal{X} = \cup_{j \in J} \mathcal{X}_j$  ( $J = \{1, \dots, N\}$ )
2: Outputs : Connected components  $C(\mathcal{X}) = \{C_1, \dots, C_k\}$ 
3: Define  $\mathcal{V} = \{\mathcal{X}_1, \dots, \mathcal{X}_N\}$ ,  $\mathcal{E} = \{\}$ 
4: For  $(i, j) \in J \times J$  do
5:   if  $\mathcal{X}_i \cap \mathcal{X}_j \neq \phi$  then
6:      $\mathcal{E} = \mathcal{E} + \{(\mathcal{X}_i, \mathcal{X}_j)\}$ 
7: Define  $k = 0$ 
8: Define  $\mathcal{H}_k = \{\}$ 
9: While  $\mathcal{V} \setminus \mathcal{H}_k \neq \phi$  do
10:   Choose  $\mathcal{X}_a \in \mathcal{V} \setminus \mathcal{H}_k$ 
11:    $k = k + 1$ 
12:    $C_k = \{\mathcal{X}_a\}$ 
13:   Define  $C' = \{\mathcal{X}_i \mid (\mathcal{X}_a, \mathcal{X}_i) \in \mathcal{E}\} \setminus C_k$ 
14:    $C_k = C_k \cup C'$ 
15:    $\mathcal{H}_k = \mathcal{H}_{k-1} \cup C_k$ 
16:   While  $C' \neq \phi$  do
17:      $C' = \{\mathcal{X}_i \mid (\mathcal{X}_i, \mathcal{X}_j) \in \mathcal{E} \text{ for some } \mathcal{X}_j \in C_k\} \setminus C_k$ 
18:      $C_k = C_k \cup C'$ 
19:      $\mathcal{H}_k = \mathcal{H}_{k-1} \cup C_k$ 
20: Return  $C(\mathcal{X}) = \{C_1, \dots, C_k\}$ 

```

Moreover, let $C(\mathcal{X}) = \{C_\ell\}_{1 \leq \ell \leq k}$ be the connected components of the graph G and let $\{J_\ell\}_{1 \leq \ell \leq k}$ be the partition of J such that $C_\ell = \{\mathcal{X}_j \mid j \in J_\ell\}$. For any ℓ ($1 \leq \ell \leq k$), put

$$T_\ell = \bigcup_{j \in J_\ell} \mathcal{X}_j (\subset \mathcal{X}), \quad \sigma_\ell = \max_{j \in J_\ell} \{\sigma(\mathcal{X}_j, \delta_j)\}.$$

Note that for any $i, j \in J_\ell$ ($1 \leq \ell \leq k$), we have that $\Sigma_i = \Sigma_j$ and $\mathbf{U}_i = \mathbf{U}_j$. Thus, we denote by $\mathbf{U}_\ell$ the matrix $\mathbf{U}_j$ for any $j \in J_\ell$ hereafter. Algorithms 3 and 4 explain the procedure to compute the connected components $C(\mathcal{X})$ of the graph $G(\mathcal{X})$ and the noise level σ_ℓ for each sentence contained in the covering $\mathcal{X} = \bigcup_{j \in J} \mathcal{X}_j$. In the computation of σ_ℓ , we can also compute the matrix $\mathbf{U}_\ell$ for each sentence as a by-product.

The construction of connected components $C(\mathcal{X})$ proceeds as follows. First, we randomly choose a vertex $\mathcal{X}_a$ from vertices $\mathcal{V}(G) = \{\mathcal{X}_j\}_{j \in J}$ and create a connected component C_1 consisting all of vertices connecting to $\mathcal{X}_a$. Next, we remove the vertices in C_1 from $\mathcal{V}(G)$ and repeat this process until all vertices in $\mathcal{V}(G)$ are removed.

Definition 6 (Covering Metric Analytical Gaussian Mechanism). For any $s \in \mathcal{X}$, let $\ell(s)$ ($1 \leq \ell(s) \leq m$) be the index such that $s \in T_{\ell(s)}$. The **Covering Metric Analytical Gaussian (CMAG)** mechanism M produces a noise-added embedding $M(s)$ for a sentence embedding s , where $M(s) = s + \mathbf{U}_{\ell(s)} \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(0, \sigma_{\ell(s)}^2 \mathbf{I}_n)$.

Theorem 4 states that CMAG is a **Covering Metric Differential Privacy (CMDP)** mechanism.

THEOREM 4. *The CMAG mechanism given by $M(s) = s + \mathbf{U}_{\ell(s)} \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(0, \sigma_{\ell(s)}^2 \mathbf{I}_n)$ satisfies $(\epsilon, \{\delta_j\}_{j \in J})$ - $d_{\mathcal{X}}$ -privacy with respect to $\{(\mathcal{X}_j, d_{\Sigma_j})\}_{j \in J}$.*

PROOF. The proof of Theorem 4 is given in Appendix A.3. □

Algorithm 5 exhibits the outline to implement CMAG for a given covering of sentences $\mathcal{X} = \bigcup_{j \in J} \mathcal{X}_j$.

ALGORITHM 4: Computation of the Matrix $\mathbf{U}_\ell$ and the Noise Level σ_ℓ for Each Sentence.

```

1: Inputs : Sentences  $\mathcal{X} = \{s_1, \dots, s_m\}$ , a covering  $\mathcal{X} = \cup_{j \in J} \mathcal{X}_j$ ,  $\varepsilon \geq 0$ ,  $\delta_j \in (0, 1)$  ( $j \in J$ )
2: Outputs : Matrices  $\mathcal{U} = \{\mathbf{U}_1, \dots, \mathbf{U}_m\}$  and standard deviations  $\text{Sd} = \{\sigma_1, \dots, \sigma_m\}$ 
3: Define  $\mathcal{U}' = \{\}$ ,  $\mathcal{U}'' = \{\}$ ,  $\text{Sd} = \{\}$ ,  $\text{Sd}' = \{\}$ 
4: Compute the connected components  $C(\mathcal{X}) = \{C_1, \dots, C_k\}$  of the graph  $G(\mathcal{X})$  by using Algorithm 3
5: For  $\ell = 1, \dots, k$  do
6:   Define  $J_\ell = \{j \in J \mid \mathcal{X}_j \in C_\ell\}$ 
7:   Define  $T_\ell = \cup_{j \in J_\ell} \mathcal{X}_j$ 
8:   Compute the sample covariance matrix  $\Sigma_\ell$  for sentences  $T_\ell$  by using Algorithm 1
9:   Compute  $\mathbf{U}_\ell = \Sigma_\ell^{1/2}$ 
10:   $\mathcal{U}' = \mathcal{U}' + \{\mathbf{U}_\ell\}$ 
11:  Define  $\text{Sd}'_\ell = \{\}$ 
12:  For  $j \in J_\ell$  do
13:    Compute  $d_{0,j} = \max\{d_{\Sigma_\ell}(s, s') \mid (s, s') \in \mathcal{X}_j^{od}\}$ 
14:    Compute the real solution  $\sigma_{0,j}$  of the equation  $f(d_{0,j}, \sigma) = \delta_j$ 
15:     $\text{Sd}'_\ell = \text{Sd}'_\ell + \{\sigma_{0,j}\}$ 
16:  Compute  $\sigma_\ell = \max \text{Sd}'_\ell$ 
17:   $\text{Sd}' = \text{Sd}' + \{\sigma_\ell\}$ 
18: For  $i = 1, \dots, m$  do
19:   Compute  $\ell(s_i)$  such that  $s_i \in T_{\ell(s_i)}$ 
20:    $\mathcal{U} = \mathcal{U} + \{\mathbf{U}_{\ell(s_i)}\}$ ,  $\text{Sd} = \text{Sd} + \{\sigma_{\ell(s_i)}\}$ 
21: Return  $\mathcal{U} = \{\mathbf{U}_{\ell(s_1)}, \dots, \mathbf{U}_{\ell(s_m)}\}$ ,  $\text{Sd} = \{\sigma_{\ell(s_1)}, \dots, \sigma_{\ell(s_m)}\}$ 

```

ALGORITHM 5: CMAG.

```

1: Inputs : Sentences  $\mathcal{X}$  with embeddings  $\mathbf{X} = \{s_1, \dots, s_m\}$ , a covering  $\mathcal{X} = \cup_{j \in J} \mathcal{X}_j$ ,  $\varepsilon \geq 0$ ,  $\delta_j \in (0, 1)$  ( $j \in J$ )
2: Outputs : Perturbed sentence embeddings  $\hat{\mathbf{X}} = \{\hat{s}_1, \dots, \hat{s}_m\}$ 
3: Define  $\hat{\mathbf{X}} = \{\}$ 
4: Compute matrices  $\mathcal{U} = \{\mathbf{U}_1, \dots, \mathbf{U}_m\}$  and standard deviations  $\text{Sd} = \{\sigma_1, \dots, \sigma_m\}$  by using Algorithm 4
5: For  $i = 1, \dots, m$  do
6:   sample  $\mathbf{z} \sim N(0, \sigma_i^2 \mathbf{I}_n)$ 
7:    $\hat{s}_i = s_i + \mathbf{U}_i \mathbf{z}$ 
8:    $\hat{\mathbf{X}} = \hat{\mathbf{X}} + \{\hat{s}_i\}$ 
9: Return  $\hat{\mathbf{X}}$ 

```

4 Experiments and Results

4.1 Settings

In this section, we use the theoretical tools developed in Section 3.2 to privatise a given set $\mathcal{X}$ of sentences. Our first step is to represent each sentence s by a fixed dimensional embedding $\mathbf{s} \in \mathbb{R}^b$. Any sentence encoder can be used for this purpose. In our experiments, we use the supervised SimCSE encoder⁶ [17], which produces 1,024-dimensional sentence embeddings. SimCSE is a contrastive learning approach for creating accurate sentence embeddings that reported excellent performance in various downstream NLP tasks requiring sentence embeddings. Moreover, follow-up work has further improved the accuracy of the sentence embeddings produced by SimCSE. We use the supervised sentence embeddings⁷ obtained by fine-tuning a RoBERTa large model on labelled

⁶<https://github.com/princeton-nlp/SimCSE>

⁷Available at <https://huggingface.co/princeton-nlp/sup-simcse-roberta-large>

sentence pairs datasets, which reported superior performance over the unsupervised sentence encoders.

We use the 1M sentences from Wikipedia, released by Gao et al. [17], in our experiments and create embeddings for those sentences as the candidate set for finding nearest neighbours for a given sentence. In order to efficiently find the top nearest neighbours of any sentence, we quantise and index the embedding vectors using SCANN [18], which is a product quantisation-based approximate nearest neighbour search method. It takes less than 10 ms to find the top-100 nearest neighbours for a given sentence among this candidate set of 1M sentences using SCANN, which is highly efficient and sufficiently accurate (reporting over 99% of recall) for our purposes.

To privatise the sentences using CMDP in a downstream evaluation task, we first create sentence embeddings for the sentences that belong to the target task using SimCSE and add those embeddings to the SCANN index we created from Wikipedia. We then find the top-100 nearest neighbours for each sentence s using SCANN and find disjunctive partitions $T_{\ell(s)}$ by merging any neighbourhoods that have common neighbours. However, due to the sparse nature of neighbourhoods in sentence embeddings, such overlaps are rare in practice. We compute the covariance matrix Σ_j from the sentence embeddings in $T_{\ell(s)}$ and add random noise according to the CMAG mechanism described in Theorem 4. Algorithm 5 shows the overall process of privatising a given set of sentence embeddings using CMAG.

4.2 Baselines and Comparisons

We compare the following DP mechanisms in our experiments.

CMAG: This is our proposed CMDP mechanism that uses the analytical Gaussian mechanism [5] to add random noise to sentence embeddings, which is given in Definition 6. As proved in Theorem 4, CMDP satisfies the definition of covering metric DP with the analytical Gaussian mechanism. For each sentence $s \in S$, let $\ell(s)$ ($1 \leq \ell(s) \leq m$) be the index such that $s \in T_{\ell(s)}$. Using the nearest neighbours of s in $T_{\ell(s)}$, we compute the corresponding covariance matrix, $\Sigma_{\ell(s)}$. We then find $\mathbf{U}_{\ell(s)}$ such that $\mathbf{U}_{\ell(s)}\mathbf{U}_{\ell(s)}^\top = \Sigma_{\ell(s)}$. Finally, we compute the noise-added embedding, $M(s)$, of s as $M(s) = s + \mathbf{U}_{\ell(s)}z$ with $z \sim \mathcal{N}(0, \sigma_{\ell(s)}^2 \mathbf{I}_n)$, as explained in Definition 6.

We experimented with two variants of CMAG in our evaluations. We use the Euclidean distance as the distance metric (corresponds to setting the covariance matrix to the identity matrix, i.e., $\mathbf{U}_{\ell(s)} = \mathbf{I}_n$) and denote this variant as **CMAG(E)**. Our proposed CMAG mechanism that uses $\mathbf{U}_{\ell(s)} = \Sigma_{\ell(s)}^{-1/2}$ is denoted by **CMAG**. By comparing CMAG(E) against CMAG, we will be able to evaluate the effect of using Euclidean vs. Mahalanobis distances; the latter uses the full covariance matrix when computing the distance between two sentences as given by Equation (3.3) (see more details below).

MAG: To understand the effect of using different metric spaces as proposed in CMDP, we use the analytical Gaussian mechanism to add random noise to all sentence embeddings, represented within the same embedding space. This corresponds to computing a single (scaled) covariance matrix, $\Sigma = \frac{n}{\text{tr}(V_X)} V_X$ from all sentence embeddings in $\mathcal{X}$ ($V_X = \frac{1}{|\mathcal{X}|-1} \sum_{s \in \mathcal{X}} (s - \bar{s})(s - \bar{s})^\top$, $\bar{s} = \frac{1}{|\mathcal{X}|} \sum_{s \in \mathcal{X}} s$). We refer to this as a MAG mechanism. By comparing the performance reported by MAG against that by the CMAG variants, we will be able to evaluate the effect of using multiple metric spaces instead of one when determining the noise level to be added to each sentence embedding.

Mahalanobis: This is the Mahalanobis mechanism proposed by Xu et al. [43], which has performed well when privatising static word embeddings. We apply this Mahalanobis mechanism to sentence embeddings, s , where the perturbed sentence embeddings, $M(s)$, are given

by, $M(s) = \mathbf{s} + \mathbf{U}\mathbf{v}$. Here, $\mathbf{U} = \Sigma^{-1/2}$ and $\mathbf{v}$ is given by Equation (4.1).

$$\mathbf{v} = y(\lambda\Sigma + (1 - \lambda)\mathbf{I}_n)^{1/2}\mathbf{x} \quad (4.1)$$

Here, $\mathbf{x} = \mathbf{z}/\|\mathbf{z}\|$ and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I}_n)$. y is sampled randomly from the Gamma distribution with shape parameter n and the scale parameter $1/\varepsilon$. Xu et al. [43] refer to the $\lambda = 1$ setting in (4.1) as the Mahalanobis mechanism. Unlike CMAG and MAG, the Mahalanobis mechanism does not estimate the noise level from the neighbourhood of each sentence.

Laplacian: Xu et al. [43] proposed the Laplacian mechanism for static word embeddings by setting $\lambda = 0$ in Equation (4.1). We apply this mechanism for the sentence embeddings in our experiments to compare the effect of considering the covariance matrix when measuring the distance between sentence embeddings.

Before we discuss the experimental results, let us present the computation and memory requirements of CMAG. First, note that in CMAG the covariance matrices are computed not over the entire set of sentences but rather per each neighbourhood. A neighbourhood typically contains 1,000 sentences on average, which are found via an efficient approximate nearest neighbour search method such as SCANN, as described earlier. Moreover, the computation of covariance matrices can be carried out in parallel independently for each neighbourhood. Second, the dimensionality of the covariance matrices is determined by the dimensionality of the sentence embeddings and not by the number of sentences in the corpus. For example, for the SimCSE sentence embeddings that we use in our experiments, we have $1,024 \times 1,024$ -dimensional covariance matrices. The total number of parameters to store across all covariance matrices (1,000 in total for a corpus of 1M sentences) is approximately 1B, which is comparatively much smaller than most LLMs that are used in NLP. Although not required for the datasets used in our experiments, one could further reduce the parameter size by performing low-rank approximations to the covariance matrices, similar to parameter-efficient optimisation methods such as LoRA. [22]

4.3 Privacy Experiments

To evaluate the level of privacy protected by CMAG against other DP mechanisms, we conduct *privacy* experiments using the following two privacy statistics that have been used in prior work [15, 43].

N_s : This is the probability of a sentence s *not* getting redacted in the mechanism (i.e., $\Pr\{M(s) = s\}$). In other words, even after adding random noise according to the mechanism M , s can be easily rediscovered. We approximate N_s by running the mechanism 100 times and counting the number of times the original (no noise added) embedding of a sentence s turns out to be the top-1 nearest neighbour of the noise-added embedding, $M(s)$, of s , where the additive noise is determined by the mechanism M .

S_s : We define $S_s = |\{s' \in \mathcal{X} : \Pr\{M(s) = s'\} \geq \eta\}|$ as the number of distinct sentences that have a probability greater than η of being the output of $M(s)$. S_s is approximated by counting the number of distinct substitutions for an input sentence s after running the mechanism 100 times. After adding noise to sentence embeddings according to a privacy mechanism M , if each sentence s has diverse neighbours, then it would become difficult to predict s from its neighbours, protecting the privacy of s , indicated by high S_s values.

Although these privacy statistics have been used in prior work for measuring the privacy in static word embeddings, to the best of our knowledge, we are the first to use those metrics for sentence embeddings. However, the two privacy statistics do not depend on the objects being considered and can be used for sentence embeddings as well. To understand the relationship between N_s , S_s , and the level of privacy guaranteed by a DP mechanism, recall that a smaller privacy budget

ϵ corresponds to a stronger privacy guarantee by adding a larger scale ($1/\epsilon$) of random noise sampled from M . This would lead to many sentence embeddings being perturbed with random noise (thus decreasing N_s), and more diverse outputs for each sentence (thus increasing S_s). Therefore, there exists an inverse relationship between N_s and S_s . For any fixed-scale noise of $1/\epsilon$, the mechanism with a better privacy guarantee will have a lower value of N_s and a higher value of S_s .

To empirically compare CMAG against previously proposed DP mechanisms, we consider the union of sentences from all benchmark datasets that are used in the utility experiments described in Section 4.4. This results in a set S of $n = 122,388$ sentences. Next, we add random noise, sampled from different mechanisms for different ϵ values. As can be inferred from the definition, N_s has a probability distribution associated with it. Different sentences s will have different N_s values. For example, the plot for the 95th percentile shows the percentage of sentences for which we can still discover the original sentence (no noise added) using the noise-added version of their sentence embeddings within the top 95% of neighbours sorted in the descending order of the probability $\Pr\{M(s) = s\}$. By reporting results at different percentile points, we can better understand the level of privacy protected by a DP mechanism rather than if we had reported for a specific setting. Figure 3 shows the 5th, 50th, and 95th percentiles of the N_s and S_s values reported by CMAG, CMAG(E), MAG, Mahalanobis, and Laplacian mechanisms for different ϵ values.

As discussed above, from Figure 3, we see that N_s values increase for all mechanisms with ϵ , while S_s values decrease. Although the relative differences among the compared mechanisms are small, especially for the N_s plots, we see that CMAG reports lower N_s values among all mechanisms for high and medium levels of noises (corresponding to smaller ϵ values) in Figures 3(a) to 3(c). Likewise, from S_s plots, we see that CMAG reports higher S_s values, especially for the 50th (Figure 3(e)) and 95th percentiles (Figure 3(f)). The Mahalanobis mechanism closely follows the performance of CMAG, while often reporting the second best level of privacy across all percentile settings. We see that CMAG(E), which uses the Euclidean distance instead of the Mahalanobis distance, performs consistently worse than CMAG. This result shows the importance of considering the full covariance matrix computed from each neighbourhood when deciding the direction and magnitude of the random noise vector that must be added to the sentence embeddings. The Laplacian mechanism reports the lowest S_s values across all percentile settings, indicating that it is not suitable for privatising sentence embeddings.

4.4 Utility Experiments

Perturbing a sentence embedding by adding random noise sampled from a particular privacy mechanism can distort the semantic information encoded in that sentence embedding. If this distortion is too strong, it could be problematic when using the noise-added sentence embeddings in downstream NLP tasks that rely on accurate sentence representations. To empirically evaluate the utility of the privatised sentence embeddings, we apply them to a broad range of NLP tasks that have been frequently used for evaluating sentence embeddings as described next in Section 4.4.1.

4.4.1 Downstream NLP Tasks. Depending on whether a task requires a single sentence or a pair of sentences as the input, we can categorise the NLP tasks into two groups. Tasks that require single-sentence representations are as follows: (1) **sentiment classification (SST)**, (2) **subjectivity status detection (SUBJ)**, (3) **question type classification (TREC)**, (4) **customer product review classification (CR)**, and (5) **movie review classification (MR)**. Tasks that require inputting a pair of sentences are as follows: (6) **natural language inference (NLI)**, (7) **paraphrase detection (MRPC)**, and (8) **semantic textual similarity (STS)** prediction. Next, we describe each of those tasks in detail and the benchmark datasets used for evaluating the performance on each task.

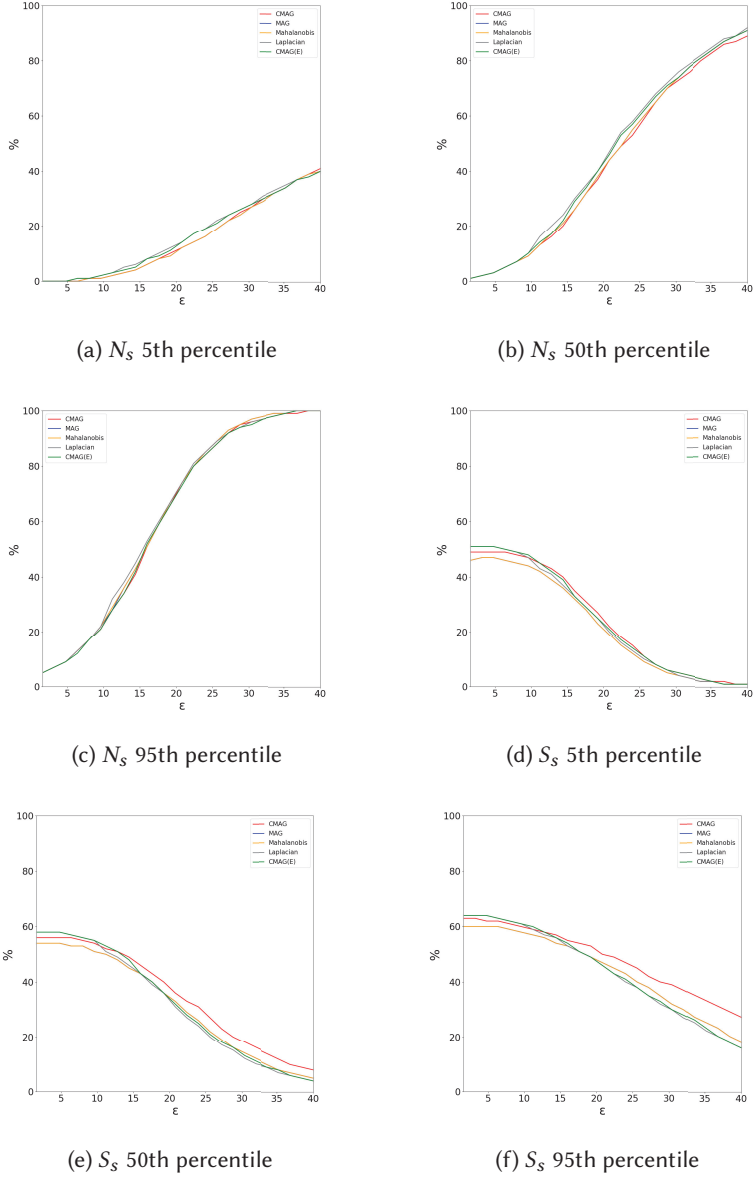


Fig. 3. N_S and S_S values reported by CMAG, CMAG(E), MAG, Mahalanobis, and Laplacian mechanisms for different ϵ values. Results are shown for the 5th, 50th, and 95th percentiles.

Sentiment Classification (SST): Socher et al. [36] introduced the Stanford Sentiment Treebank, a sentence-level sentiment treebank containing sentences annotated for sentiment at various levels in the dependency parse tree. Following the standard practice, we conduct sentence-level (at the root of the dependency tree) binary (positive vs. negative) sentiment classification using this dataset. The dataset contains 67K training sentences and 1.8K testing sentences. We first create a sentence embedding for each sentence using SimCSE and then

add random noise according to a privacy mechanism under evaluation. We experiment with different privacy mechanisms and at different privacy parameters ϵ as discussed later. Next, we use the SentEval [12]⁸ sentence embedding evaluation tool to train a binary classifier using the training sentences in the benchmark datasets for each task. The binary classification accuracy is reported as the evaluation metric in the subsequent experiments. A privacy mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Subjectivity Status Detection (SUBJ): A sentence might express a subjective or an objective opinion, whereas sentiment polarity (i.e., positive vs. negative sentiment) must only be predicted for subjective opinions. Pang and Lee [32] showed that by first training a binary classifier for predicting whether a given sentence expresses a subjective or an objective opinion and then classifying only those sentences labelled as subjective opinions for sentiment can improve the accuracy of sentiment classification. Using 5,000 movie reviews, they released a subjectivity dataset in which sentences are associated with binary (subjective vs. objective) labels for this purpose. This dataset contains 10K training sentences and 10K testing sentences. We use SimCSE to represent sentences as 1,024-dimensional embeddings, and add noise according to different privacy mechanisms. We then use the SentEval tool to train a binary classifier using the noise-added sentence embeddings. The binary classification accuracy is reported as the evaluation metric in the subsequent experiments. A privacy mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Question Type Classification (TREC): To correctly answer free-form factual questions given a large collection of texts, one must first understand the type of information that should be included in the answer to the question. For example, given the question “*What Canadian city has the largest population?*”, we would classify it having the answer type **city**, implying that only cities are valid as the candidate answers. Li and Roth [27] created a dataset from four question datasets, including TREC collections, and manually labelled into a hierarchical taxonomy of question types. Each question is classified into the following six top-level types: ABBREVIATION, ENTITY, DESCRIPTION, HUMAN, LOCATION, and NUMERIC.⁹ We use SimCSE to represent sentences as 1,024-dimensional embeddings, and add noise according to different privacy mechanisms. We then use the SentEval tool to train a multi-class classifier using the noise-added sentence embeddings for the sentences in the train split of this dataset. The macro-averaged accuracy is reported as the evaluation metric in the subsequent experiments. A privacy mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Customer Product Review Classification (CR): Hu and Liu [23] compiled a dataset of customer reviews from e-commerce websites to evaluate the sentiment polarity expressed by the customers on the products sold on those sites. This dataset has been used extensively as a benchmark for evaluating the performance of sentiment classifiers [39]. Each sentence from a customer review is assigned either a positive or a negative sentiment label. There are 4K sentences each in the training and testing splits of this dataset. Similar to the SST setting above, we use SimCSE to represent sentences as 1,024-dimensional embeddings and add noise according to different privacy mechanisms. We then use the SentEval tool to train a binary classifier using the noise-added sentence embeddings. The binary classification accuracy is reported as the evaluation metric in the subsequent experiments. A privacy

⁸<https://github.com/facebookresearch/SentEval/tree/main>

⁹Details of the full hierarchy are available at <https://cogcomp.seas.upenn.edu/Data/QA/QC/definition.html>

mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Movie Review Classification (MR): The movie review dataset [39] contains sentences selected from reviews written on movies and are selected from the IMDB. Each sentence is assigned a binary sentiment label (i.e., positive vs. negative). There are 11K training sentences and 11K testing sentences in this dataset. Similar to the SST setting above, we use SimCSE to represent sentences as 1,024-dimensional embeddings and add noise according to different privacy mechanisms. We then use the SentEval tool to train a binary classifier using the noise-added sentence embeddings. The binary classification accuracy is reported as the evaluation metric in the subsequent experiments. A privacy mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Natural Language Inference (NLI): Given a premise and a hypothesis expressed by two sentences, in NLI, we are expected to predict whether the hypothesis has an entailment, contradiction, or neutral relation with the premise. For example, given the premise “*A small girl wearing a pink jacket is riding on a carousel*” and the hypothesis “*The carousel is moving*”, there exists an entailment relationship between the premise and the hypothesis. Unlike the NLP tasks discussed above, NLI prediction requires us to represent each sentence in a pair of sentences and make a prediction for the pair as a whole instead of individual sentences in the pair. We use the SICK Entailment dataset [30] for evaluating the performance of sentence embeddings for NLI. This dataset has 4.5K training instances and 4.9K testing instances. First, we create sentence embeddings for each sentence in a premise–hypothesis pair using SimCSE. Next, we add noise according to different privacy mechanisms. Finally, we concatenate the two sentence embeddings to represent a sentence pair and use the SentEval tool to train a ternary classifier. The macro-averaged classification accuracy is reported as the evaluation metric in the subsequent experiments. A privacy mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Paraphrase Detection (MRPC): In this task, we are required to predict whether two given sentences are paraphrases (i.e., express highly similar meanings). The Microsoft Research Paraphrase Corpus (MRPC) [13] is a popular dataset that has been used for evaluating the ability of sentence embedding methods to detect paraphrases. This dataset contains sentence pairs extracted from news sources on the Web. The sentence pairs are annotated by humans, indicating whether they capture a paraphrase/semantic equivalent relationship. For example, the two sentences “*The procedure is generally performed in the second or third trimester*” and “*The technique is used during the second, and occasionally, third trimester of pregnancy*” are considered as paraphrases in the MRPC dataset. This is a binary classification task over a pair of sentences. The MRPC dataset contains 4.1K training instances and 1.7K testing instances. Following the same procedure described above for NLI, we train a binary classifier (instead of the ternary classifier trained for entailment) using instances in MRPC. The classification accuracy is reported as the evaluation metric in the subsequent experiments. A privacy mechanism that retains the information of the original sentences will report high accuracy for this task, indicating high utility.

Semantic Textual Similarity (STS): The STS task considers the semantic similarity between two given sentences. In STS datasets, each sentence pair is rated by a group of human annotators from an ordinal rating 0 to 5, where 0 indicates that the two sentences in a pair are completely dissimilar (e.g., “*The black dog is running through the snow.*” vs. “*A race car driver is driving his car through the mud.*”), whereas 1 indicates that the two sentences are completely equivalent (i.e., mean the same thing, e.g., “*The bird is bathing in the sink.*” vs.

“*Birdie is washing itself in the water basin.*”). STSBenchmark [10] comprises a selection of the English datasets used in the STS tasks organised in the context of the SemEval task series between 2012 and 2017. It contains sentences selected from diverse genres such as image captions, news headlines, and user forums.

STS is considered as a task for evaluating sentence embeddings. For example, all sentences are represented by a sentence embedding and their similarity is computed as the cosine of the angle between the two embeddings. The correlation between the computed cosine similarity scores and the human ratings is used as the evaluation metric in STS, and the Spearman correlation coefficient is used for this purpose. Spearman correlation coefficient is between -1 and $+1$; a high positive correlation is considered as an indication of an accurate sentence encoder.

In all utility experiments for which a classifier is required, we train a linear classifier to minimise cross-entropy loss. RMSProp [37] is used as the optimiser with a batch size of 128 instances. We iterate twice over the training datasets for each task (i.e., maximum number of epochs is set to 2).

4.4.2 Results for the Utility Experiments. For each utility task described in Section 4.4.1, we evaluate the performance reported by our proposed DP mechanism, CMAG, against the baseline methods MAG, Mahalanobis, Laplacian, and CMAG(E). These methods were already introduced in Section 4.2. We generate random noise from each of those mechanisms under different privacy levels indicated by ϵ to privatise the SimCSE sentence embeddings for the sentences in the benchmark dataset associated with each task. For ease of description, in Figure 4, we show results for the tasks in which we add random noise to individual sentences, whereas Figure 5 shows results for the tasks in which we add random noise to pairs of sentences.

From Figure 4, we see that for sentiment classification (SST; shown in Figure 6(a)) our proposed CMAG consistently outperforms all other DP mechanisms across almost all values of ϵ , reporting high classification accuracies. Recall that smaller ϵ values indicate stronger privacy guarantees, which requires us to add high levels of random noise (e.g., corresponding to high standard deviations under the Gaussian mechanism). Therefore, for smaller ϵ values, we see that all DP mechanisms report worse performance, which improves with ϵ . Laplacian and CMAG(E) report the second best performance for SST, and it is difficult to separate the two corresponding curves. The fact that CMAG outperforms CMAG(E) shows that it is important to consider the full covariance when measuring the distance between neighbours in the neighbourhood of a sentence instead of using the Euclidean distance. The worst performing DP mechanisms are Mahalanobis and MAG; their curves overlap for the most part in Figure 6(a). Note that MAG uses a single metric space when determining the random noise to be added to a sentence embedding, unlike CMAG, which uses a different metric space for each neighbourhood. The superior performance reported by CMAG over MAG shows the importance of using different metric spaces as proposed by our CMDP setting. This finding supports our claim that the neighbourhood densities of sentence embeddings differ significantly from one sentence to another, which must be taken into account when determining the level of random noise that must be added to a particular sentence to protect its privacy.

Although the Laplacian mechanism shows better utility here, its privacy protection is rather minimal, as previously discussed in Section 4.3, which is a concern when using Laplacian as a DP mechanism for sentence embeddings. The Mahalanobis mechanism that was originally proposed by Xu et al. [43] as a DP mechanism for static word embeddings does not perform well when applied to sentence embeddings. Compared with static word embeddings, sentence embeddings are high dimensional. For example, word2vec [31] or GloVe [33] embeddings are 300 dimensional, whereas the sentence embeddings produced by SimCSE are available at 768 (base) or 1,024 (large) dimensionalities. Moreover, a sentence contains multiple words, which would require different levels of privacy protection than in the case of a single word. For example, functional words such

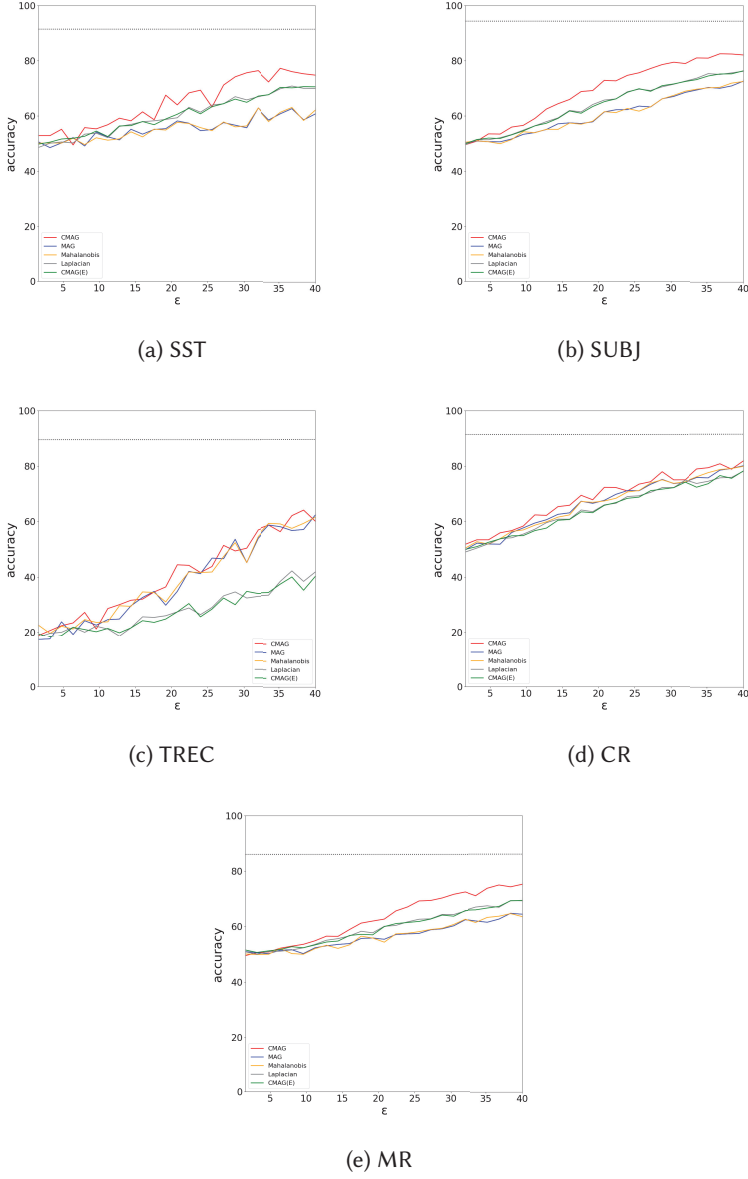


Fig. 4. Utility tasks in which random noise is added by a DP mechanism to individual sentences. The sub-figures show the performance reported for the following tasks: sentiment classification (SST) (a), subjectivity status detection (SUBJ) (b), question type classification (TREC) (c), customer product review classification (CR) (d), and movie review classification (MR) (e). The dotted line in each sub-figure shows the level of performance we would obtain if no noise was added to the sentence embeddings (i.e., using the non—privacy protected, original sentence embeddings).

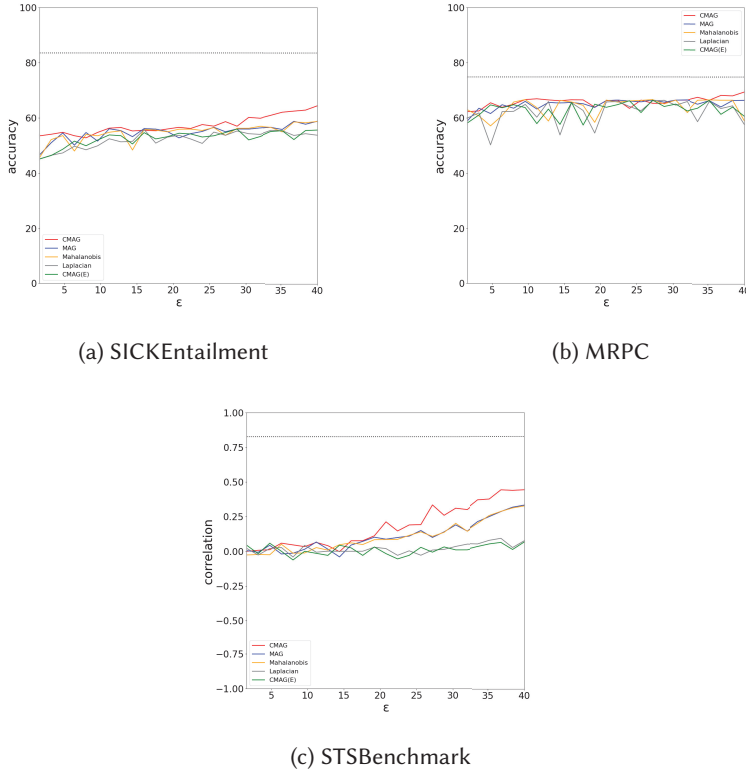


Fig. 5. Utility tasks in which random noise is added by a DP mechanism to each of the two sentences in sentence-pair instances. The sub-figures show the performance reported for the following tasks: natural language inference (NLI) (a), paraphrase detection (MRPC) (b), and semantic textual similarity (STS) (c). The dotted line in each sub-figure shows the level of performance we would obtain if no noise was added to the sentence embeddings (i.e., using the non-privacy protected, original sentence embeddings).

as articles (also known as *stop* words) do not contain sensitive information and would not require high random noise to be added, whereas content words such as named entities (e.g., personal names, locations, and organisations) could be carrying sensitive information that would need to be protected by adding a stronger level of noise. We believe that these important differences between word versus sentence embeddings must be carefully considered when designing DP mechanisms for sentence embeddings and, as we see here, direct application of DP mechanisms that have shown good performance for word embeddings (such as the Mahalanobis mechanism) does not perform well for sentence embeddings.

We see similar trends as the ones discussed above for SUBJ (Figure 6(b)), CR (Figure 6(d)), and MR (Figure 6(e)), which are all sentiment- or subjectivity-related sentence-level prediction tasks. On TREC (Figure 6(c)), we see again MAG and Mahalanobis mechanisms reporting similar performances, which is better than that reported by CMAG(E) and Laplacian mechanisms. The question classification task in TREC is more complex (involves six top-level categories in a hierarchy) than sentiment/subjectivity classification tasks (mostly binary classification tasks).

From Figure 5, we see that even for the tasks that require considering a pair of sentences as the input, the proposed CMAG reports the best performance across different privacy levels, specified

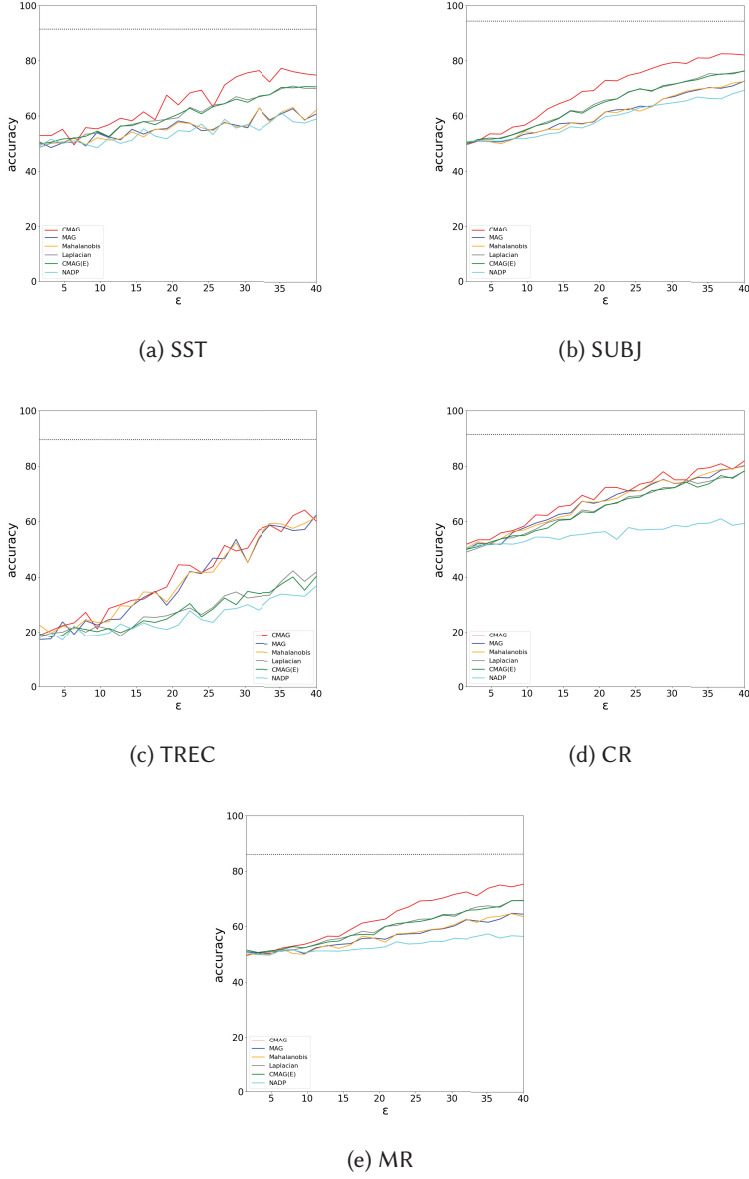


Fig. 6. Comparing NADP, a word-level DP mechanism applied to sentences against sentence-level DP mechanisms. The sub-figures show the performance reported for the following tasks: sentiment classification (SST) (a), subjectivity status detection (SUBJ) (b), question type classification (TREC) (c), customer product review classification (CR) (d), and movie review classification (MR) (e). The dotted line in each sub-figure shows the level of performance we would obtain if no noise was added to the sentence embeddings (i.e., using the non-privacy protected, original sentence embeddings).

by ϵ . However, the relative differences in performance among CMAG(E), MAG, Mahalanobis, and Laplacian are less obvious in SICKEntailment (Figure 5(a)) and MRPC (Figure 5(b)) tasks. Considering that each sentence in an input instance is independently privatised by additive random noise, the total noise inserted in pairwise tasks such as NLI, MRPC, and STS (shown in Figure 5) are relatively higher than that in single-sentence tasks (shown in Figure 4). On the STSBenchmark (Figure 5(c)), we see that for $\epsilon > 20$, CMAG reports consistently higher Spearman correlation coefficients than all other methods.

In all experiments, we see that the differences among all DP mechanisms in the strong privacy (low ϵ) regime is indistinguishable, however. Therefore, an important future research direction in sentence-level DP mechanisms would be to develop methods that guarantee high utility in the strong privacy regimes.

4.5 Comparisons against Word-Level Differential Privacy Mechanisms

Given that much prior work on DP has focused on word-level mechanisms that add noise to individual word embeddings, an interesting question from an application point of view is how well a word-level DP mechanism would perform when used to privatise sentence embeddings. To empirically study this, we extend the current state-of-the-art word-level differential privacy method, NADP [8] to sentence embeddings and compare CMAG against this method in the revised version. Specifically, we use the same text encoder (i.e., SimCSE) as CMAG for NADP, where we represent each word w_i in a sentence s by a word embedding $\mathbf{w}_i$. The word embeddings are obtained by passing that word as the sole input to the SimCSE encoder. Next, NADP is used to add random noise to those word embeddings, $\text{NADP}(\mathbf{w}_i)$. Finally, we average the noise-added word embeddings to obtain private sentence embedding $\mathbf{s} = \frac{1}{|s|} \sum_{\mathbf{w}_i \in s} \text{NADP}(\mathbf{w}_i)$. Here, $|s|$ denotes the number of words in the sentence s . By dividing the sentence embedding by the length of the sentence (i.e., the number of words in the sentence), we prevent longer sentences accumulating a higher noise. Because our goal here is to evaluate the effect of privacy protection in sentence-level tasks, we use the SST, SUBJ, TREC, CR, and MR benchmarks described above, which evaluate the accuracy of sentence embeddings. The accuracies obtained at different privacy levels ϵ are shown in Figure 6 for NADP and the sentence-level DP mechanisms that were already introduced above.

From Figure 6, we see that NADP reports the worst performance in almost all sentence-level benchmarks, whereas CMAG reports the best performance across different privacy levels. This result shows that simply applying word-level DP mechanisms to add random noise to individual words is *not* the optimal strategy for privatising sentence embeddings. We believe that there are two reasons for this. First, word-level DP mechanisms consider each word independently of the other words in the same sentence when determining the level of random noise that must be added to a given word embedding. The set of words in a sentence is not a random collection of objects; rather, together those words express a coherent meaning at sentence level. Therefore, it is more appropriate to use the sentence embedding that encodes the information conveyed by the entire sentence rather than adding noise to individual words in the sentence. Second, because random noise is added to each word separately, the total noise added to the entire sentence could become too high, thus distorting the semantic information included in the final sentence embedding, computed as the average of the noise-added word embeddings. This is especially problematic for long sentences containing many words. Although we divide the sentence embedding by the number of words in that sentence to address this sentence length-related bias, this alone seems to be inadequate to mitigate this sentence length bias, as evident from the experimental results. Therefore, we conclude that DP mechanisms defined at sentence level, such as our proposed CMAG, must be used to privatise sentence embeddings instead of applying word-level DP mechanisms.

Table 1. Nearest Neighbours of Sentences Retrieved at Different Noise Levels for CMAG and Laplacian DP Mechanisms

Noise Level	Mechanism	Nearest Neighbour
<i>On the morning of December 6, 2019, a terrorist attack occurred at Naval Air Station Pensacola in Pensacola, Florida.</i>		
No noise		On 27 December 2019, the K-1 Air Base in Iraq, which hosts Iraqi and U.S. personnel, was attacked, killing an American contractor.
Medium	CMAG	The heats were started on 24 August at 11:20.
	Laplace	The city is a large industrial center and is served by Tambov Donskoye Airport.
High	CMAG	The Round of 16 matches were held on 15 and 16 April 1982.
	Laplace	The EASR suite's initial per-unit cost will be about \$180 million less than the DBR, for which the estimate is about \$500 million.
<i>Hatam Mohammed Alir al-Sami (June 5, 1951 – September 14, 2011) was an Iranian terrorist, mostly known for being a planner of the September 11 attacks, along with Osama Bin Laden.</i>		
No noise		Zein Isa was a member of the Abu Nidal Organization, which at the time he murdered his daughter, was plotting to bomb the Israeli Embassy in Washington, D.C.. Harris wrote that "His daughters had long worried about Zein's ANO activities.
Medium	CMAG	During the early Muslim conquests in the 630s, al-Ubulla was conquered by the Arab forces of Utba ibn Ghazwan al-Mazini after the defeat of its 500-man Sasanian garrison.
	Laplace	In 2012, he was indicted by the U.S. District Court for the District of Columbia for drug trafficking activities.
High	CMAG	Both battalions attacked with over in waves of two companies each, at Without reconnaissance, the battalions ran into obstacles halfway to their objective.
	Laplace	In the next match, Konosuke Takeshita faced Hiroshi Tanahashi from New Japan Pro-Wrestling.
<i>His research team has been supported by grants from the National Institute of General Medical Sciences and the National Cancer Institute to design and examine novel therapeutic strategies for patients with kidney, bladder and breast cancer.</i>		
No noise		The main areas of his research during this period are liver resection, liver and kidney transplantation, laparoscopic and metabolic surgery, stem cells and genetic studies.
Medium	CMAG	His research approaches integrate computational modeling, cell biology, mouse models, and genetic and biochemical experimentation to understand metabolic processes and their contribution to health.
	Laplace	Others have taken place at festivals, cafes, empty warehouses, and the largest yet took place on 4 December 2018 at the Woodbrooke Quaker Centre in Birmingham had 45 participants from 12 different countries and ranging from 4 to 70 years of age.
High	CMAG	All research results will be presented both to the scientific community and to the general public.
	Laplace	At the Bureau of Public Enterprises, BPE Aziz was a Deputy Director in charge of Information Technology saddled with the task of formulating the agency's Information Technology strategy, frameworks and implementation for a successful overhaul of government's privatisation policy.

4.6 A Case Study of Nearest Neighbours

We conduct a qualitative case study to understand the effect of adding noise to sentence embeddings using CMAG. For a given sentence, we add random noise sampled from CMAG and find the top nearest neighbouring sentence from a set of 1M sentences randomly selected from the English-language version of Wikipedia. If the nearest neighbour sentence turns out to be highly semantically similar to the original sentence, the DP mechanism would not protect the user privacy adequately. We conduct this analysis for medium and high levels of noise (determined by the privacy parameter ϵ) for CMAG and compare the retrieved top nearest neighbours against that for the Laplacian DP mechanism.

In Table 1, we have shown the nearest neighbours for three sentences: the top two sentences about terrorism activities and the last one about a medical research project. Without adding any

random noise, we see that sentences highly similar to the original sentence are retrieved as its nearest neighbours, indicating that it is not possible to protect the sensitive information in a sentence simply by replacing it with nearest neighbours. However, when we add medium-level noise with CMAG, we see that we obtain sentences that are topically relevant to the original sentence but do not disclose the sensitive information such as the exact dates and named entities mentioned in the original sentence. This is important from the point of view of protecting user privacy while retaining sufficient information to conduct any subsequent NLP tasks. On the other hand, it is quite difficult to strike a fine balance between privacy and utility with the Laplacian DP mechanism. For example, the Laplacian DP mechanism returns totally out-of-topic sentences or ones that are highly similar to the original sentence.

5 Conclusion

In this article, we proposed a metric DP method for protecting sensitive information in sentences when they are used in downstream NLP tasks. Our proposed CMAG method determines the strength of random noise that must be added to the embedding of a sentence, computed from a pretrained sentence encoder, considering the nearest neighbours of that sentence. Theoretical analysis proves that CMAG satisfies CMDP requirements, thus providing theoretically proven privacy guarantees. Moreover, results from the empirical privacy experiments support the theoretical claims, in which CMAG protects information encoded in sentence embeddings. Given that sentence embeddings are used as the input in a wide range of NLP tasks, we evaluate the utility of the noise-added sentence embeddings on a battery of standard benchmark datasets that are frequently used to evaluate sentence embeddings. Results from the utility experiments show that the privatised sentence embedding using the proposed CMAG outperformed all other DP mechanisms when applied to sentence embeddings. Although we considered sentence embeddings in this article, the theoretical analysis we presented does not impose any limitations on the length of the sentences being considered.

Given that LLMs are arguably the most popular application of NLP, it is worth considering how our proposed CMAG could be applied to protect the privacy of the users using LLM-powered tools such as ChatGPT and Gemini. In such chat-based interfaces, billions of users engage in detailed conversations with LLM-based agents to describe their needs and obtain relevant information. Some of these conversations could use sensitive information about the users, such as their medical conditions or social relationships, that the users would not wish to be used for purposes other than providing relevant information for themselves. CMAG could potentially be used in this setting to protect user privacy by first encoding the input text from the user using a suitable encoder (we can use LLM-based embeddings for this purpose) and then use CMAG to add noise to the encoded input. Only the noise-added embeddings would be sent to the LLM service. The suitable level of noise that strikes a balance between utility and privacy must be carefully determined considering the generations given by the LLM service. Recall that CMAG requires a set of sentences to be given from which we are selecting the nearest neighbours for a given input sentence to be privatised. In particular, the input sentence *does not* need to be given in advance in this setting. If the set of sentences from which we select nearest neighbours is sufficiently large enough to cover a broad range of topics, then there will be some neighbourhood that can cover any given input sentence. Exploring the extensions of CMAG to LLMs is beyond the scope of the current work and would be an interesting future research direction.

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Appendix

A Proofs of Theorems

A.1 Proof of Theorem 2

Let $p_{M(s)}(y) = p_{M(s)}(y_1, \dots, y_n)$ be probability density of the distribution of the random variable $Y_s = M(s)$, that is, we have that

$$\Pr \{Y_s \in E\} = \Pr \{M(s) \in E\} = \int_E p_{M(s)}(y)$$

for any measurable set E . Then, the privacy loss function $\ell_{M,s,s'}(y)$ and the privacy loss random variable $L_{M,s,s'}$ of M on a pair $s, s' \in \mathcal{X}$ are defined as follows (see Balle and Wang [5, Section 2.2]):

$$\ell_{M,s,s'}(y) = \log \left(\frac{p_{M(s)}(y)}{p_{M(s')}(y)} \right), \quad L_{M,s,s'} = \ell_{M,s,s'}(Y_s).$$

For simplicity, let us put

$$p(y) = p_{M(s)}(y), \quad p'(y) = p_{M(s')}(y), \quad Y = Y_s, \quad Y' = Y_{s'}.$$

LEMMA 1. *The privacy loss random variables $L_{M,s,s'}$ and $L_{M,s',s}$ follow the same distribution $\mathcal{N}(\eta, 2\eta)$ with $\eta = d_\Sigma(s, s')^2 / 2\sigma^2$.*

Since, by symmetricity, $L_{M,s,s'}$ and $L_{M,s',s}$ follow the same distribution, it is enough to prove that $L_{M,s,s'} \sim \mathcal{N}(\eta, 2\eta)$. In this proof, we use the next well-known property of random variables:

PROPOSITION 1. *Let $X = (X_1, \dots, X_n)^\top$ be an n -dimensional random variable, $\mu \in \mathbb{R}^n$ and let V be an $n \times n$ positive definite matrix. Then, the next properties (a) and (b) are equivalent:*

- (a) X follows the normal distribution $\mathcal{N}(\mu, V)$.
- (b) For any $m \times n$ matrix M and any $v \in \mathbb{R}^m$, $MX + v$ follows the normal distribution $\mathcal{N}(M\mu + v, MVM^\top)$.

By Proposition 1 and the assumption $z \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$, we have that

$$\begin{aligned} Y = Y_s &= s + \mathbf{U}z \sim \mathcal{N}(s, \sigma^2 \mathbf{U}\mathbf{U}^\top) = \mathcal{N}(s, \sigma^2 \Sigma), \\ Y' = Y_{s'} &= s' + \mathbf{U}z \sim \mathcal{N}(s', \sigma^2 \mathbf{U}\mathbf{U}^\top) = \mathcal{N}(s', \sigma^2 \Sigma), \end{aligned}$$

which implies that

$$\begin{aligned} p(y) &= \frac{1}{(2\pi)^{\frac{n}{2}} |\sigma^2 \Sigma|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} (y - s)^\top \Sigma^{-1} (y - s) \right\}, \\ p'(y) &= \frac{1}{(2\pi)^{\frac{n}{2}} |\sigma^2 \Sigma|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} (y - s')^\top \Sigma^{-1} (y - s') \right\}. \end{aligned}$$

Therefore,

$$\begin{aligned}
 L_{M,s,s'} &= \log \left(\frac{p(Y)}{p'(Y)} \right) = \frac{1}{2\sigma^2} \{ (Y - s')^\top \Sigma^{-1} (Y - s') - (Y - s)^\top \Sigma^{-1} (Y - s) \} \\
 &= \frac{1}{2\sigma^2} \{ (s + \mathbf{U}z - s')^\top \Sigma^{-1} (s + \mathbf{U}z - s') - (\mathbf{U}z)^\top \Sigma^{-1} \mathbf{U}z \} \\
 &= \frac{1}{2\sigma^2} \{ (s - s')^\top (\mathbf{U}^2)^{-1} \mathbf{U}z + (\mathbf{U}z)^\top (\mathbf{U}^2)^{-1} (s - s') + (s - s')^\top \Sigma^{-1} (s - s') \} \\
 &= \frac{1}{2\sigma^2} \{ (s - s')^\top \mathbf{U}^{-1} z + z^\top \mathbf{U}^{-1} (s - s') + (s - s')^\top \Sigma^{-1} (s - s') \}.
 \end{aligned}$$

Here, note that $(s - s')^\top \mathbf{U}^{-1} z$ is a 1-dimensional random variable. Hence,

$$z^\top \mathbf{U}^{-1} (s - s') = \{ (s - s')^\top \mathbf{U}^{-1} z \}^\top = (s - s')^\top \mathbf{U}^{-1} z.$$

Thus, we have that

$$\begin{aligned}
 L_{M,s,s'} &= \frac{1}{2\sigma^2} \{ (s - s')^\top \mathbf{U}^{-1} z + z^\top \mathbf{U}^{-1} (s - s') + (s - s')^\top \Sigma^{-1} (s - s') \} \\
 &= \frac{1}{\sigma^2} (s - s')^\top \mathbf{U}^{-1} z + \frac{1}{2\sigma^2} (s - s')^\top \Sigma^{-1} (s - s').
 \end{aligned}$$

Hence, by Proposition 1 and the assumption that $z \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$, we have that

$$\begin{aligned}
 L_{M,s,s'} &\sim \mathcal{N} \left(\frac{1}{2\sigma^2} (s - s')^\top \Sigma^{-1} (s - s'), \sigma^2 \cdot \frac{1}{\sigma^2} (s - s')^\top \mathbf{U}^{-1} \cdot \frac{1}{\sigma^2} \mathbf{U}^{-1} (s - s') \right) \\
 &= \mathcal{N} \left(\frac{d_\Sigma(s, s')^2}{2\sigma^2}, \frac{d_\Sigma(s, s')^2}{\sigma^2} \right) = \mathcal{N}(\eta, 2\eta),
 \end{aligned}$$

which completes the proof of Lemma 1.

LEMMA 2. *The mechanism M satisfies (ϵ, δ) - d_X -privacy if and only if the following holds for any $s, s' \in \mathcal{X}$:*

$$\Pr \{ L_{M,s,s'} \geq \epsilon d_\Sigma(s, s') \} - e^{\epsilon d_\Sigma(s, s')} \Pr \{ L_{M,s',s} \leq -\epsilon d_\Sigma(s, s') \} \leq \delta.$$

PROOF. By using probability density, we can rewrite (3.2) as follows:

$$\int_E \left(p(y) - e^{\epsilon d(s, s')} p'(y) \right) \leq \delta, \quad (\text{A.1})$$

where $d(s, s') = d_\Sigma(s, s')$ in this case. Put $q(y) = p(y) - e^{\epsilon d(s, s')} p'(y)$ and $P = \{y \mid q(y) \geq 0\}$, $N = \{y \mid q(y) < 0\}$. Then, we can estimate the left-hand side of (A.1) as follows:

$$\begin{aligned}
 \int_E \left(p(y) - e^{\epsilon d(s, s')} p'(y) \right) &= \int_{E \cap P} q(y) + \int_{E \cap N} q(y) \leq \int_{E \cap P} q(y) \\
 &\leq \int_P q(y) = \int_P \left(p(y) - e^{\epsilon d(s, s')} p'(y) \right).
 \end{aligned}$$

Since E is arbitrary, we can conclude that the mechanism M satisfies (ϵ, δ) - d_X -privacy if and only if

$$\int_P \left(p(y) - e^{\epsilon d(s, s')} p'(y) \right) \leq \delta \quad (\text{A.2})$$

holds for any $s, s' \in \mathcal{X}$. However, we have that

$$\begin{aligned} \Pr \{L_{M,s,s'} \geq \varepsilon d(s, s')\} &= \Pr \left\{ \log \left(\frac{p(Y)}{p'(Y)} \right) \geq \varepsilon d(s, s') \right\} \\ &= \Pr \left\{ p(Y) \geq e^{\varepsilon d(s, s')} p'(Y) \right\} = \Pr \{Y \in P\} = \int_P p(y) \end{aligned} \quad (\text{A.3})$$

and

$$\begin{aligned} \Pr \{L_{M,s',s} \leq -\varepsilon d(s, s')\} &= \Pr \left\{ \log \left(\frac{p'(Y')}{p(Y')} \right) \leq -\varepsilon d(s, s') \right\} \\ &= \Pr \left\{ \log \left(\frac{p(Y')}{p'(Y')} \right) \geq \varepsilon d(s, s') \right\} = \Pr \{Y' \in P\} = \int_P p'(y). \end{aligned} \quad (\text{A.4})$$

Hence, by (A.2), (A.3), and (A.4), we get the claim. $\square$

LEMMA 3. *The following holds for any $\varepsilon \geq 0$:*

$$\begin{aligned} \Pr \{L_{M,s,s'} \geq \varepsilon d_\Sigma(s, s')\} &= \Phi \left(\frac{d_\Sigma(s, s')}{2\sigma} - \varepsilon\sigma \right), \\ \Pr \{L_{M,s',s} \leq -\varepsilon d_\Sigma(s, s')\} &= \Phi \left(-\frac{d_\Sigma(s, s')}{2\sigma} - \varepsilon\sigma \right). \end{aligned}$$

PROOF. Put $\eta = d_\Sigma(s, s')^2 / 2\sigma^2$. By Lemma 1 and the well-known property $\Pr \{\mathcal{N}(0, 1) \geq t\} = \Pr \{\mathcal{N}(0, 1) \leq -t\}$,

$$\begin{aligned} \Pr \{L_{M,s,s'} \geq \varepsilon d_\Sigma(s, s')\} &= \Pr \{\mathcal{N}(\eta, 2\eta) \geq \varepsilon d_\Sigma(s, s')\} \\ &= \Pr \left\{ \sqrt{2\eta} \cdot \mathcal{N}(0, 1) + \eta \geq \varepsilon d_\Sigma(s, s') \right\} = \Pr \left\{ \mathcal{N}(0, 1) \geq \frac{-\eta + \varepsilon d_\Sigma(s, s')}{\sqrt{2\eta}} \right\} \\ &= \Pr \left\{ \mathcal{N}(0, 1) \leq \frac{\eta - \varepsilon d_\Sigma(s, s')}{\sqrt{2\eta}} \right\} = \Phi \left(\frac{\eta - \varepsilon d_\Sigma(s, s')}{\sqrt{2\eta}} \right) = \Phi \left(\frac{d_\Sigma(s, s')}{2\sigma} - \varepsilon\sigma \right) \end{aligned}$$

and

$$\begin{aligned} \Pr \{L_{M,s',s} \leq -\varepsilon d_\Sigma(s, s')\} &= \Pr \{\mathcal{N}(\eta, 2\eta) \leq -\varepsilon d_\Sigma(s, s')\} \\ &= \Pr \left\{ \sqrt{2\eta} \cdot \mathcal{N}(0, 1) + \eta \leq -\varepsilon d_\Sigma(s, s') \right\} = \Pr \left\{ \mathcal{N}(0, 1) \leq \frac{-\eta - \varepsilon d_\Sigma(s, s')}{\sqrt{2\eta}} \right\} \\ &= \Phi \left(\frac{-\eta - \varepsilon d_\Sigma(s, s')}{\sqrt{2\eta}} \right) = \Phi \left(-\frac{d_\Sigma(s, s')}{2\sigma} - \varepsilon\sigma \right), \end{aligned}$$

which completes the proof of Lemma 3. $\square$

Theorem 2 is a direct consequence of Lemma 2 and Lemma 3.

A.2 Proof of Theorem 3

(1), (2) Put

$$m(d, \sigma) := \frac{d}{2\sigma} - \varepsilon\sigma, \quad n(d, \sigma) := -\frac{d}{2\sigma} - \varepsilon\sigma.$$

Then, since

$$\frac{\partial}{\partial \sigma} m(d, \sigma) = -\frac{d}{2\sigma^2} - \varepsilon, \quad \frac{\partial}{\partial \sigma} n(d, \sigma) = \frac{d}{2\sigma^2} - \varepsilon, \quad \exp \left(-\frac{m(d, \sigma)^2}{2} \right) = \exp(\varepsilon d) \exp \left(-\frac{n(d, \sigma)^2}{2} \right),$$

we have that

$$\begin{aligned}
 f_\sigma(d, \sigma) &= \frac{\partial}{\partial \sigma} f(d, \sigma) = \frac{\partial}{\partial \sigma} \left\{ \Phi(m(d, \sigma)) - e^{\varepsilon d} \Phi(n(d, \sigma)) \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \left\{ \frac{\partial m}{\partial \sigma} \cdot \exp\left(-\frac{m(d, \sigma)^2}{2}\right) - \frac{\partial n}{\partial \sigma} \cdot \exp(\varepsilon d) \cdot \exp\left(-\frac{n(d, \sigma)^2}{2}\right) \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \left\{ \frac{\partial m}{\partial \sigma} \cdot \exp\left(-\frac{m(d, \sigma)^2}{2}\right) - \frac{\partial n}{\partial \sigma} \cdot \exp\left(-\frac{m(d, \sigma)^2}{2}\right) \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{m(d, \sigma)^2}{2}\right) \cdot \left(-\frac{d}{\sigma^2}\right) < 0.
 \end{aligned}$$

Moreover, since

$$\lim_{\sigma \rightarrow +0} m(d, \sigma) = \infty, \quad \lim_{\sigma \rightarrow +0} n(d, \sigma) = -\infty, \quad \lim_{\sigma \rightarrow \infty} m(d, \sigma) = -\infty, \quad \lim_{\sigma \rightarrow \infty} n(d, \sigma) = -\infty,$$

we have that

$$\begin{aligned}
 \lim_{\sigma \rightarrow +0} f(d, \sigma) &= \lim_{\sigma \rightarrow +0} \left\{ \Phi(m(d, \sigma)) - e^{\varepsilon d} \Phi(n(d, \sigma)) \right\} = 1 - e^{\varepsilon d} \cdot 0 = 1, \\
 \lim_{\sigma \rightarrow \infty} f(d, \sigma) &= \lim_{\sigma \rightarrow \infty} \left\{ \Phi(m(d, \sigma)) - e^{\varepsilon d} \Phi(n(d, \sigma)) \right\} = 0 - e^{\varepsilon d} \cdot 0 = 0.
 \end{aligned}$$

Therefore, the function $f_{s,s'}(\sigma) = f(d_\Sigma(s, s'), \sigma)$ ($\sigma > 0$) is strictly decreasing from 1 to 0 on $(0, \infty)$, which implies that the equation $f_{s,s'}(\sigma) = \delta$ ($\delta \in (0, 1)$) has the unique real solution $\sigma(s, s', \delta) = \inf\{\sigma \mid f_{s,s'}(\sigma) \leq \delta\}$.

(3) Let (s, s') be any element of $\mathcal{X}^{od}$. By definition, we have that $\sigma(s, s', \delta) \leq \sigma_0$. Then, as shown in the proof of Theorem 3 (1), we have that $f_{s,s'}(\sigma) \leq \delta$ for any $\sigma \geq \sigma(s, s', \delta)$. In particular, we have that $f_{s,s'}(\sigma_0) \leq \delta$, which, by Theorem 2, implies that the mechanism satisfies (ε, δ) - d_X -privacy.

(4) We first prove the next lemma.

LEMMA 4. *The equation $F(d, \sigma) = f(d, \sigma) - \delta = 0$ determines an implicit function $\sigma = g(d)$ for any $d > 0$, which satisfies $\frac{dg}{dd} > 0$.*

PROOF. As shown in the proof of Theorem 3 (1), we have that

$$\frac{\partial}{\partial \sigma} F(d, \sigma) = f_\sigma(d, \sigma) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{m(d, \sigma)^2}{2}\right) \cdot \left(-\frac{d}{\sigma^2}\right) < 0 \quad (d > 0),$$

which, by the implicit function theorem, implies that the equation $F(d, \sigma) = 0$ defines an implicit function $\sigma = g(d)$ and $dg/dd = -F_d/F_\sigma$ for any $d > 0$. Moreover, since

$$\frac{\partial}{\partial d} m(d, \sigma) = \frac{1}{2\sigma}, \quad \frac{\partial}{\partial d} n(d, \sigma) = -\frac{1}{2\sigma},$$

we have that

$$\begin{aligned}
\frac{\partial}{\partial d} F(d, \sigma) &= f_d(d, \sigma) \\
&= \frac{1}{\sqrt{2\pi}} \left\{ \left(\frac{1}{2\sigma} \right) \cdot \exp\left(-\frac{m(d, \sigma)^2}{2}\right) - \left(-\frac{1}{2\sigma}\right) \cdot \exp(\varepsilon d) \cdot \exp\left(-\frac{n(d, \sigma)^2}{2}\right) \right\} \\
&\quad - \varepsilon \cdot \exp(\varepsilon d) \cdot \Phi(n(d, \sigma)) \\
&= \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp(\varepsilon d) \cdot \exp\left(-\frac{n(d, \sigma)^2}{2}\right) - \varepsilon \cdot \exp(\varepsilon d) \cdot \Phi(n(d, \sigma)) \\
&= \exp(\varepsilon d) \left\{ \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left(-\frac{n(d, \sigma)^2}{2}\right) - \varepsilon \cdot \Phi(n(d, \sigma)) \right\}. \tag{A.5}
\end{aligned}$$

Note that, the error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

satisfies the relation

$$\Phi(x) = \frac{1}{2} \left\{ 1 + \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right) \right\} = \frac{1}{2} \left\{ 1 - \operatorname{erf}\left(-\frac{x}{\sqrt{2}}\right) \right\} \tag{A.6}$$

and the inequality

$$\frac{1}{x + \sqrt{x^2 + 2}} < e^{x^2} \int_x^\infty e^{-t^2} dt \leq \frac{1}{x + \sqrt{x^2 + \frac{4}{\pi}}} \quad (x \geq 0)$$

holds (Abramowitz and Stegun [45]). In particular, we have that

$$e^{x^2} \int_x^\infty e^{-t^2} dt < \frac{1}{x + \sqrt{x^2}} = \frac{1}{2x} \quad (x > 0). \tag{A.7}$$

Then, since $\int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$, we have that

$$1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} dt - \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt. \tag{A.8}$$

Thus, by (A.6), (A.8), we have that

$$\Phi(n(d, \sigma)) = \frac{1}{2} \left\{ 1 - \operatorname{erf}\left(-\frac{n(d, \sigma)}{\sqrt{2}}\right) \right\} = \frac{1}{\sqrt{\pi}} \int_{-\frac{n(d, \sigma)}{\sqrt{2}}}^\infty e^{-t^2} dt.$$

Hence, by using (A.7), we can estimate the underlined part of (A.5) as follows;

$$\begin{aligned}
& \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left(-\frac{n(d, \sigma)^2}{2}\right) - \varepsilon \cdot \Phi(n(d, \sigma)) \\
&= \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left(-\frac{n(d, \sigma)^2}{2}\right) - \frac{\varepsilon}{\sqrt{\pi}} \int_{-\frac{n(d, \sigma)}{\sqrt{2}}}^{\infty} e^{-t^2} dt \\
&= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{n(d, \sigma)^2}{2}\right) \left\{ \frac{1}{\sigma} - \sqrt{2}\varepsilon \cdot \exp\left(\frac{n(d, \sigma)^2}{2}\right) \int_{-\frac{n(d, \sigma)}{\sqrt{2}}}^{\infty} e^{-t^2} dt \right\} \\
&> \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{n(d, \sigma)^2}{2}\right) \left\{ \frac{1}{\sigma} - \sqrt{2}\varepsilon \cdot \frac{1}{-\sqrt{2}n(d, \sigma)} \right\} \\
&= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{n(d, \sigma)^2}{2}\right) \cdot \frac{\frac{d}{\sqrt{2}\sigma} + \sqrt{2}\varepsilon\sigma - \sqrt{2}\varepsilon\sigma}{-\sqrt{2}n(d, \sigma)\sigma} \left(\because n(d, \sigma) = -\frac{d}{2\sigma} - \varepsilon\sigma \right) \\
&= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{n(d, \sigma)^2}{2}\right) \cdot \frac{d}{-2n(d, \sigma)\sigma^2} > 0.
\end{aligned}$$

Therefore, we have that $F_{\sigma}(d, \sigma) < 0$ and $F_d(d, \sigma) > 0$. Then, by the implicit function theorem, we can conclude that

$$\frac{dg}{dd} = -\frac{F_d(d, \sigma)}{F_{\sigma}(d, \sigma)} > 0,$$

which completes the proof. $\square$

Let (s, s') be any element of $\mathcal{X}^{od}$. By Theorem 3 (2), $d_{\Sigma}(s, s')$ and $\sigma(s, s', \delta)$ satisfy $F(d_{\Sigma}(s, s'), \sigma(s, s', \delta)) = 0$ or $\sigma(s, s', \delta) = g(d_{\Sigma}(s, s'))$. By Lemma 4, the function $g(d)$ is monotone increasing, which implies that $\sigma_0 = \max\{\sigma(s, s', \delta) \mid (s, s') \in \mathcal{X}^{od}\}$ is attained by the most distant pair (s_0, s'_0) .

A.3 Proof of Theorem 4

Let $j \in J$ be an arbitrary index and let (s_*, s'_*) be any element of $\mathcal{X}_j^{od}$. Suppose that $j \in J_{\ell}$. Then, for any $s \in \mathcal{X}_j$, we have that $s \in T_{\ell}$, which implies that $\ell(s) = \ell$, $\sigma_{\ell(s)} = \sigma_{\ell}$ and $\mathbf{U}_{\ell(s)} = \mathbf{U}_{\ell} = \mathbf{U}_j$. Since $\sigma(\mathcal{X}_j, \delta_j) = \max\{\sigma(s, s', \delta_j) \mid (s, s') \in \mathcal{X}_j^{od}\}$ and $j \in J_{\ell}$, we have that

$$\sigma(s_*, s'_*, \delta_j) \leq \sigma(\mathcal{X}_j, \delta_j) \leq \sigma_{\ell}. \quad (\text{A.9})$$

Moreover, by Theorem 3 (1), (2), we have that

$$f_{s_*, s'_*}(\sigma) \leq \delta_j \quad (\sigma \geq \sigma(s_*, s'_*, \delta_j)). \quad (\text{A.10})$$

Hence, by (A.9) and (A.10), we have that $f_{s_*, s'_*}(\sigma_{\ell}) \leq \delta_j$. Therefore, by applying Theorem 2 to the metric space $(\mathcal{X}_j, d_{\Sigma_j})$, we can conclude that the Gaussian output perturbation mechanism $M(s) = s + \mathbf{U}_{\ell(s)}z$ with $z \sim \mathcal{N}(0, \sigma_{\ell(s)}^2 \mathbf{I}_n)$ satisfies (ε, δ_j) - $d_{\mathcal{X}_j}$ -privacy. That is, we have that

$$\Pr\{M(s) \in E\} \leq e^{\varepsilon d_{\Sigma_j}(s, s')} \Pr\{M(s') \in E\} + \delta_j$$

for any $s, s' \in \mathcal{X}_j$ and any measurable set $E \subset \mathbb{R}^n$. Since $j \in J$ is arbitrary, we get the claim.

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